

CORRELATION

4.1 Introduction

After understanding the basics of statistics, we have already studied how to collect, classify, tabulate, and present data in a meaningful way. We also learned about important measures of central tendency like mean, median, and mode, which help to summarise data into a single representative value. Now, in this chapter, we move one step further in business data analysis by studying two important topics – **Correlation** and **Regression**. These methods help us to understand how two or more variables are related and how one variable can be predicted by knowing the value of another.

? What is Correlation?

The term correlation is used by a common man without knowing the term correlation. Correlation simply means a relationship or connection between two things. When one thing changes, and another thing also changes, we say there is correlation between them.

Everyday Examples of Correlation

1. Study and Exam Marks

- When parents say, "Study well to get a good job," they are actually pointing to a correlation.
- More study hours → Better exam marks → Higher chances of getting a good job
- This means, as one thing increases, the other also increases. This is an example of positive correlation.

2. Rainfall and Crop Growth

- When we say, "More rainfall helps crops grow better," we are showing a relationship between rainfall and crop yield.
- More rainfall → More water in soil → Better crop production
- This again is a positive correlation because both things move in the same direction.

3. Watching TV and Study Time

- A student who spends more time watching TV usually spends less time studying.
- More TV time → Less study time → Lower academic performance
- This is an example of negative correlation, where one value increases and the other decreases.

In statistics, **Correlation** refers to the relationship between two or more variables. It shows how the values of one variable are related to the values of another. Correlation helps us to measure how strong or weak the relationship is between two things (called variables). These variables can be anything that can change — like income, temperature, sales, age, marks, etc.

The word "**relationship**" is important because it highlights that there is some kind of link between the variables. However, correlation does not prove that one variable causes the other to change; it simply shows that they are related. For instance, price and supply, or income and expenditure, can be correlated, but one may not necessarily cause the other.

Scenario	Correlated Variables
A student studies more and scores higher	Study Time & Exam Marks
A shopkeeper sees fewer buyers when prices go up	Price & Demand
Older people tend to have higher blood pressure	Age & Blood Pressure
Experienced workers perform better	Experience & Output

Correlation is a useful tool in statistics that helps to understand whether the relationship between two variables is positive (both increase or decrease together) or negative (one increases while the other decreases), and how strong or weak that relationship is. When a change in one variable is matched by a change in another, the two variables are said to be correlated. For example, price and demand, or height and weight, show a relationship and are considered correlated. Correlation shows how one variable behaves when the other changes. It explains the relationship between two variables, usually denoted as x and y . This relationship is measured using a statistical tool called the coefficient of correlation, which is a number that ranges from $+1$ to -1 . In the next section, we will begin by understanding correlation analysis, which is used to study and measure the strength of relationships.

The study of statistics can be done in different ways depending on the number of variables involved.

1. **Univariate Analysis** : When we study only one variable, like height, weight, age, marks, wages, it is called Univariate Analysis. This type of study tells about the basic features of a single set of data.
2. **Bi-Variate Analysis**. When we study the relationship between two variables, it is called Bivariate Analysis. For example, in health studies, we may look at the relationship between age and blood pressure, or between nutrient intake and weight gain, or between income and medical expenses. In such cases, the variables are often connected to each other. To understand how strong this connection is, we use **Correlation** and **Regression Analysis**.

4.1.1

Meaning and Definition of Correlation



Meaning: Correlation

Correlation refers to the relationship between two or more variables. It shows how the values of one variable are related to the values of another. It is sometimes termed as "Covariation". The measure of correlation is called **co-efficient of correlation**.

Definitions: Correlation

1. According to A.M. Tuttle, "An analysis of the relationship of two or more variables is usually called Correlation."
2. According to Croxton and Cowden, "When the relationship is of a quantitative nature, the appropriate statistical tool for discovering and measuring the relationship and expressing it in a brief formula is known as Correlation."
3. According to L.R. Connor, "If two or more quantities vary in sympathy so that movements in one tend to be accompanied by corresponding movements in other(s) then they are said to be correlated."
4. According to Ya Lun Chou, "Correlation analysis attempts to determine the 'degree of relationship' between variables".

4.1.2 Importance and Uses of Correlation in Data Analysis

1. Measures the Degree of Relationship Between Variables

Correlation helps to understand how strongly two numerical variables like income and expenditure, price and supply, or supply and demand are related.

2. Estimates the Value of an Unknown Variable

If the value of one variable is known, correlation can be used to predict or estimate the value of the other variable.

3. Identifies Key Influencing Variables

It helps in finding out which variables are important and have an impact on other variables in the data.

4. Supports Accurate Predictions

By knowing the strength of the relationship, it becomes easier to make reliable and exact predictions in real-life situations.

5. Helps in Calculating Sampling Error

Correlation is useful in measuring how much variation or error exists in sample data when compared to the full population.

6. Forms the Base for Regression Analysis

Correlation is the foundation for understanding regression, which is used to predict one variable based on another.

7. Helps in Business and Economic Decision Making

Correlation helps managers, economists, and planners to understand the link between market factors such as demand and pricing, helping them to make better and informed decisions.

8. Useful in Scientific and Social Research

In fields like medicine, psychology, and education, correlation helps in examining the strength of relationships between factors, such as medicine dosage and recovery time, or study hours and academic performance.

4.2 Types of Correlation

Correlation can be studied and classified in different ways based on the nature of the relationship between variables. The type of correlation depends on whether the variables move in the same or opposite direction, whether the relationship is regular or irregular, and how many variables are being studied. These classifications help in understanding the direction, pattern, and complexity of the relationship. The three main ways to classify correlation are:

4.2.1 Types of Correlation Based on Direction of Change

This type of classification is based on whether the two variables increase or decrease together or in opposite directions. Whether the correlation between the variables is positive or negative depends on its direction of change.

1. **Positive or Direct Correlation:** The correlation is positive when both the variables move in the same direction. If values of variables are increasing or decreasing in the same direction, then such correlation is called as positive correlation.

Examples Positive or Direct Correlation

- **Height and Weight :** When a person's height increases, their weight also increases. Both go up together, showing positive correlation.
- **Study Time and Exam Marks :** When study time increases, exam marks also increase. Both rise together, so this is positive correlation.
- **Physical Exercise and Stress Level :** When a person reduces physical exercise, their ability to manage stress also decreases. Both go down together, so this is a case of positive correlation even though both values are decreasing.

2. **Negative or Inverse Correlation:** The correlation is negative when the two variables move in opposite directions. If the value of one variable increases and the other decreases, it is called as negative correlation.

Examples Negative or Inverse Correlation

- **Price and Demand:** When the price of a product increases, its demand decreases. They move in opposite directions. This shows negative correlation.
- **Temperature and Use of Woolen Clothes:** As the temperature increases, the use of woolen clothes decreases. One goes up while the other goes down, which is negative correlation.
- **Stress Level and Concentration:** When stress increases, a person's ability to concentrate decreases. These two move in opposite directions, so this is a negative correlation.

4.2.2 Types of Correlation Based on the Type of Relationship (Form of Change)

This classification depends on whether the relationship between the variables follows a constant or irregular pattern of change.

1. **Linear Correlation:** Linear correlation means that both variables change at a regular or steady rate. If one variable increases or decreases by a fixed amount, the other also changes by a fixed amount. On a graph, the points form a straight line.

Examples Linear Correlation

- There is a 10-unit increase in variable 'x', and at the same time, there is a 10-unit increase in variable 'y'. This shows that both variables are increasing by the same amount, which is a sign of linear correlation. The change is regular and follows a fixed pattern.

x	10	20	30	40	50
y	30	40	50	60	70

- **Study Hours and Test Scores:** For every 1 extra hour of study, test marks increase by 5. The increase is steady.
- **Years of Service and Salary:** For every 1 year of experience, salary increase by Rs.5000. The increase is steady.

2. **Non-linear Correlation:** Non-linear correlation means the change is not regular. As one variable changes, the other may increase or decrease, but not in a fixed way. On a graph, the points form a curve or irregular pattern.

Examples Non-linear Correlation

- The changes in x do not lead to a regular or fixed pattern in y. Sometimes y increases, sometimes it decreases, and the amount of change is not constant. This irregular pattern means the relationship between x and y is non-linear.

x	7	8	9	10	11
y	6	11	17	13	5

- **Study Time and Memory:** Studying helps memory, but too much studying without rest can reduce understanding.
- **Temperature and Crop Yield:** A little more heat may help crops grow, but too much heat can reduce the yield.

4.2.3 Types of Correlation Based on the Number of Variables Studied

This classification is based on how many variables are studied at the same time.

1. **Simple Correlation:** It involves the study of the relationship between two variables only.

Examples Simple Correlation

These are examples of simple correlation, where the relationship between only two variables is studied at a time. In each case, we observe how one variable affects the other, without involving any third factor.

- **Income and Expenditure:** As income increases, spending also increases.
- **Hours of Sleep and Alertness:** Less sleep leads to lower alertness.
- **Rainfall and Crop Growth:** More rainfall generally leads to better crop production.

2. **Partial Correlation:** In this, the relationship between two variables is studied while keeping the effect of other variables constant.

Examples Partial Correlation

These are examples of partial correlation, where the relationship between two main variables is studied, but the effect of one or more other variables is kept constant or ignored. This helps in understanding the direct link between two variables without interference from outside factors.

- **Study Time and Exam Marks (Keeping Health Constant):** A student who studies more usually scores higher, but to focus only on study time and marks, the impact of health is kept constant.
- **Exercise and Weight Loss (Keeping Diet Constant):** More exercise usually leads to weight loss, but in this case, we assume the diet remains the same to study the pure effect of exercise on weight.
- **Training and Job Performance (Keeping Experience Constant):** Better training can improve job performance, and by keeping work experience constant, we focus only on the impact of training.

3. **Multiple Correlation:** This involves the study of three or more variables at the same time to see how they are related together.

Examples Multiple Correlation

These are examples of multiple correlation, where the relationship among three or more variables is studied together. The goal is to see how a combination of variables influences another variable.

- **Income, Education, and Spending:** A person's spending habits may depend on both their income level and their educational background. Studying all three together shows how income and education together affect spending.
- **Exercise, Diet, and Health Condition:** A person's health may depend not only on exercise but also on diet. Studying both together helps to understand the combined effect on health.
- **Experience, Technical Skills, and Salary:** A person's salary may increase not just with work experience, but also with better skills. Studying both helps understand what contributes most to earnings.

4.3 What is Correlation Coefficient

The **correlation coefficient** is a numerical value that shows the strength and direction of the relationship between two variables. It is represented by the symbol " r " and its value always lies between -1 and $+1$.

This value helps us understand whether the two variables move together, move in opposite directions, or have no relationship at all. It shows both the strength and direction of the relationship. It is widely used in business, economics, science, and research to study connections between different factors.

1. $r = +1$ (Perfect Positive Correlation)

This means the two variables increase or decrease together in a perfect and consistent manner. The points on the graph form a straight upward line.

Example: If a person studies more, marks increase every time in the same way.

2. $r = -1$ (Perfect Negative Correlation)

This means one variable increases while the other decreases in a perfect pattern. The graph will show a straight downward line.

Example: If the price of a product increases, demand decreases regularly in the same way.

3. $r = 0$ (No Correlation)

This means there is no link between the two variables. Changes in one variable do not affect the other in any fixed way. The graph will have scattered points with no pattern.

Example: Shoe size and exam marks — completely unrelated.

4.4 What is Degree of Correlation?

The degree of correlation explains how closely the two variables are related. It tells whether the relationship is perfect, strong, weak, or absent.

1. $r = +1$ or -1 (Perfect Correlation)

This means the relationship between the two variables is exact and consistent. All points lie on a straight line, either rising or falling.

- **Perfect Positive ($r = +1$):** Both variables increase or decrease together at the same rate.

Example: Study time and marks increase exactly in the same way.

- **Perfect Negative ($r = -1$):** One variable increases, the other decreases exactly at the same rate.

Example: Price goes up, and demand falls in perfect proportion.

2. r is Close to $+1$ or -1 (High Degree of Correlation/Strong Correlation)

When the value of r is near $+1$ or -1 , it means there is a strong correlation between the variables. They are not perfectly related, but the dots on the graph are close to a straight line, either going upward (positive) or downward (negative). This shows that the variables move together in a consistent way, either in the same or opposite direction.

Example: As advertising increases, sales also rise steadily — this shows a strong positive correlation.

3. r is Close to 0 (Low Degree of Correlation/Weak Correlation)

When the value of r is near 0, it means the variables have a weak correlation. The dots are widely scattered, but they may still show a slight upward or downward pattern. The connection exists, but it is not strong or reliable.

Example: A small link between sleep hours and productivity — the relationship is present but weak.

4. $r = 0$ (No Correlation / Absent Correlation)

When r equals 0, there is no relationship between the two variables. The dots on the graph are scattered randomly with no clear pattern. Changes in one variable do not affect the other at all. This is known as absent correlation.

Example: Shoe size and exam marks — these are completely unrelated.

Summary Table

Correlation Coefficient (r)	Meaning	Graph Pattern
+1	Perfect positive correlation	Straight upward line
-1	Perfect negative correlation	Straight downward line
0	No correlation	Random scattered points
Close to +1 or -1	High Degree of Correlation	Dots close to a line
Close to 0	Low Degree of Correlation	Dots scattered with a weak trend

4.5 Methods of Measuring Correlation

The correlation can be measured by any of the following methods:

1. Scatter Diagram Method.
2. Karl Pearson's Coefficient of Correlation.
3. Spearman's Rank Correlation Coefficient

4.5.1 Scatter Diagram Method

The scatter diagram method is the simplest and most visual way to study the relationship between two variables. In this method, each pair of values (such as income and spending, or study time and marks) is represented as a dot on a graph. If there are 10 observations, then there will be 10 dots on the graph.

How it Works

- One variable is marked along the X-axis (horizontal line).
- The other variable is marked along the Y-axis (vertical line).
- Each pair of values is plotted as a dot on the graph.
- After plotting all points, we look at the overall pattern to understand how closely the two variables are related.

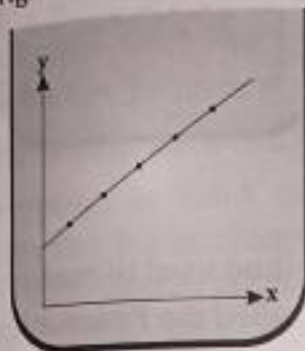
Understanding the Scatter Pattern:

- If the dots form a clear line or direction, the correlation is strong.
- If the dots are loosely scattered, the correlation is weak.
- If the dots show no pattern and are spread randomly, then there is no correlation.

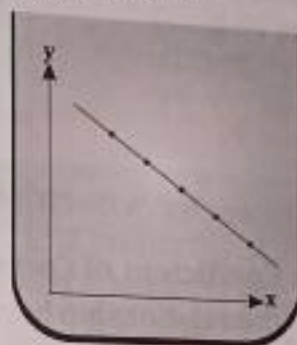
The closer the dots are to an imaginary straight line, the stronger the relationship. The more spread out the dots, the weaker the relationship. This visual method helps in quickly identifying the direction and strength of the correlation.

The following types of scatter diagrams tell about the degree of correlation between variable X and variable Y.

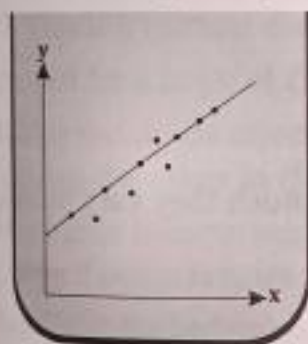
1. **Perfect Positive Correlation ($r=+1$):** The correlation is said to be perfectly positive when all the points lie on the straight line rising from the lower left-hand corner to the upper right-hand corner.



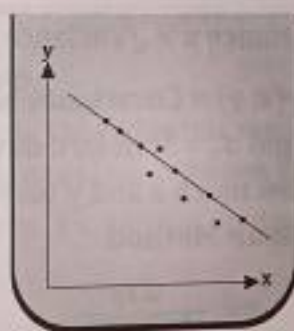
2. **Perfect Negative Correlation ($r=-1$):** When all the points lie on a straight line falling from the upper left-hand corner to the lower right-hand corner, the variables are said to be negatively correlated.



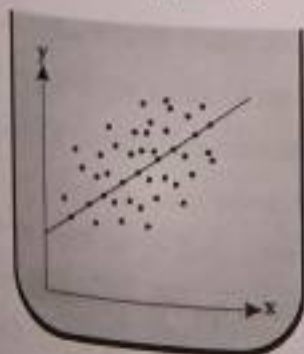
3. **High Degree of Positive Correlation ($r=+High$):** The degree of correlation is high when the points plotted fall under the narrow band and is said to be positive when these show the rising tendency from the lower left-hand corner to the upper right-hand corner.



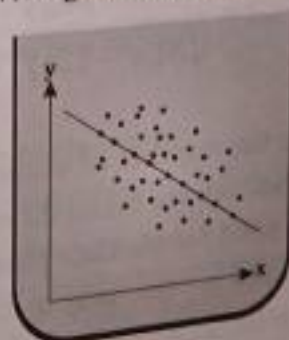
4. **High Degree of Negative Correlation ($r=-High$):** The degree of negative correlation is high when the point plotted fall in the narrow band and show the declining tendency from the upper left-hand corner to the lower right-hand corner.



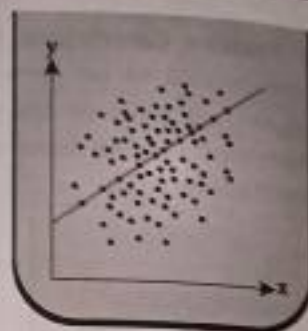
5. **Low degree of Positive Correlation ($r=+Low$):** The correlation between the variables is said to be low but positive when the points are highly scattered over the graph and show a rising tendency from the lower left-hand corner to the upper right-hand corner.



6. **Low Degree of Negative Correlation ($r = - Low$):** The degree of correlation is low and negative when the points are scattered over the graph and the show the falling tendency from the upper left-hand corner to the lower right-hand corner.



7. **No Correlation ($r = 0$):** The variable is said to be unrelated when the points are haphazardly scattered over the graph and do not show any specific pattern. Here the correlation is absent and hence $r = 0$.



4.5.2 Karl Pearson's Coefficient of Correlation

Karl Pearson's Coefficient of Correlation is a mathematical method used to measure the strength and direction of the relationship between two variables. It is also called the **Pearsonian Correlation Coefficient** and is denoted by the symbol ' r '. This method helps to find out how closely two variables (x and y) are related, and whether the relationship is positive or negative.

Three methods are used to calculate Karl Pearson's Coefficient of Correlation (r):

1. Covariance Method (General Formula)

This method uses covariance and standard deviation of the two variables.

$$r = \frac{\text{Co-variance}(x,y)}{\sqrt{\text{Variance } x} \times \sqrt{\text{Variance } y}} \text{ or } \frac{\text{Cov}(x,y)}{\sigma_x \cdot \sigma_y}$$

- $\text{Cov}(x, y)$ = Covariance between x and y
- σ_x and σ_y = Standard deviations of x and y

It tells how much x and y vary together, compared to how much they vary individually.

2. Actual Mean Method

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \times \sqrt{\sum y^2}}$$

Where, $x = X - \bar{x}$; $y = Y - \bar{y}$

This method gives an accurate result but needs more steps, especially if the values are large.

3. Direct Method :

This is a shortcut method, used when the mean is not given and values are raw.

$$r = \frac{N\sum xy - \sum x \cdot \sum y}{\sqrt{N\sum x^2 - (\sum x)^2} \times \sqrt{N\sum y^2 - (\sum y)^2}}$$

Where:

- N is the number of observations
- All values are taken directly from the dataset
- No need to calculate mean separately

This method is especially useful in competitive exams or when quick calculation is needed.

Each of these methods gives the same result (value of r) but are chosen based on what data is available and how easy the calculation is.

Assumptions

1. **Variab**
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Assumptions of Karl Pearson's Coefficient of Correlation

1. Variables Must be Numerical (Quantitative)

The two variables (x and y) should be measured using numbers, like marks, income, height, weight, price, etc. They should be on a ratio scale or interval scale, not categories like male/female or colors.

Example: If studying the relationship between income and expenditure, both values are measured in rupees and are numerical.

2. The Relationship Between Variables should be Linear

The correlation is valid only when the relationship between the variables is linear. This means that if x increases or decreases, y also changes in a proportional and steady way. On a graph, the plotted points should form a straight line pattern, not a curve.

Example: If study hours increase regularly and marks increase in the same pattern, the relationship is linear.

3. The Data should Follow a Normal Distribution

The individual values of x and y should be spread in a natural pattern, where most values lie near the average, and few values are very low or very high. This forms a bell-shaped curve when drawn as a graph and is known as **normal distribution**.

Example: In a class, most students may score between 60–70 marks. A few score below 40 or above 90. This is a normal distribution of marks.

4. There should be a Logical Cause-and-Effect Relationship

The variables should be affected by some underlying causes. This does not mean one variable directly causes the other to change, but there should be a logical link between them.

Example: Higher income may lead to more spending. There is a natural connection between the two, even if other factors are involved.

5. Variables Must Be Independent of Each Other

Each variable should change on its own, and one should not be a fixed result of the other. This helps ensure that the correlation shows a real relationship, not a forced one.

Example: Weight and exercise are independent. A person's weight may be influenced by diet, activity, and other habits, not just one factor.



Merits of Karl Pearson's Coefficient of Correlation

- Shows both the strength and direction of the relationship between two variables.
- Helps to predict the value of one variable if the other is known.
- The value of r always lies between -1 and $+1$, which makes it easy to interpret.
- It is widely used and accepted in fields like business, economics, and research.
- Works well with large data sets and gives accurate results when assumptions are met.



Demerits of Karl Pearson's Coefficient of Correlation

- Assumes the relationship between variables is linear, which may not always be the case.
- Takes more time to calculate than simpler methods like scatter diagrams.
- Sensitive to extreme values, which may give misleading results.
- Needs careful interpretation, as people may wrongly assume causation from correlation.
- Not suitable for qualitative data (like color, brand, or gender), as it only works with numerical variables.

Interpretation of 'r' - Karl Pearson's Coefficient of Correlation

1. If $r = 0$, then there is no relationship between variables.
2. If $r = +1$, then there exists perfect positive correlation between variables.
3. If $r = -1$, then there exists perfect negative correlation between variables.
4. If $r > 0.75$ but < 1 , then there is high degree of positive correlation.
5. If $r > -0.75$ but < -1 , there exist high degree of negative correlation.
6. If $r > +0.5$ but < 0.75 , then there exist moderate degree of positive correlation.
7. If $r > -0.5$ but < -0.75 , then there exist moderate degree of negative correlation.
8. If $r > 0$ but < 0.5 , then there exist low degree of positive correlation.



Note

Positive or negative relationship depends on the signs (+ or -). The coefficient of correlation measures not only the magnitude of correlation but also tells the direction. Such as, $r = -0.57$, which shows correlation is negative because the sign is "-" and the magnitude is 0.57.

Probable Error of Coefficient of Correlation [PE(r)]

Probable Error is a small value used to test whether the correlation coefficient (r) is strong and reliable or not. It helps to check whether the value of r (which shows the relationship between two variables) is just by chance or it is actually meaningful.

Suppose we calculate $r = 0.5$ from a sample of data. We want to know: Is this value of 0.5 a real correlation, or did it happen by chance because of a small or unusual sample? Here, Probable Error gives us a range or limit to check this.

If the correlation is not strong enough compared to PE, it means the result is not reliable. But if the value is much bigger than the PE, it means the result is definitely meaningful.



Meaning: Probable Error

The probable error of coefficient of correlation is an amount which if added to and subtracted from the value of r gives two limits within which coefficient of correlation obtained will fall. It is 0.6745 times of the standard error of r . It is computed as.

Standard Error

Probable Error

Where,

Interpretation

- If $|r| < 0.6$
 - If $|r| > 0.6$
- It shows which the give a range



Note :

Always use the only tells the direction

Uses of Probable Error

- a. It is used to test whether the correlation is reliable or not.
- b. It helps to check whether the value of r is just by chance or it is actually meaningful.
- c. It helps to check whether the value of r is just by chance or it is actually meaningful.
- d. It is used to test whether the correlation is reliable or not.

Probable Error

Illustration

In a bivariate correlation, $r = 0.5$ and $n = 49$

Solution

Standard Error

Standard Error

$$\text{Standard Error (SE)} = \frac{1-r^2}{\sqrt{n}}$$

$$\text{Probable Error (PE}(r)) = 0.6745 \times \frac{1-r^2}{\sqrt{n}}$$

Where, r = Coefficient of Correlation, n = Number of pairs of Observation.

Interpretation of Probable Error

- If $|r| < 6 \times \text{PE} \rightarrow$ The correlation is not significant and it may be due to chance.
- If $|r| > 6 \times \text{PE} \rightarrow$ The correlation is significant.

It shows a real relationship between the variables. In this case, we calculate the limits within which the true value of correlation may lie. The limits are $r - \text{PE}$ and $r + \text{PE}$. These two values give a range showing where the true population correlation is likely to fall.



Note :

Always use the absolute value of r ($|r|$) when comparing with $6 \times \text{PE}$, because the sign (positive or negative) only tells the direction of the relationship, not its strength.

Uses of Probable Error

- It is used to find the limits within which the true population correlation coefficient is likely to lie.
- It helps to check whether the sample correlation is statistically significant or just due to chance.
- It helps in interpreting the reliability of the correlation result, especially in small samples.
- It is used in statistical testing to decide whether to accept or reject a correlation as meaningful.

Problems on Karl Pearson's Coefficient of Correlation

Illustration 1

In a bivariate data on x and y , $\text{Var}(x) = 9$, $\text{Var}(y) = 49$ and $\text{Cov}(x, y) = 20$ find r .

Solution

$$\text{Var}(y) = 49, \text{Var}(x) = 9, \text{Cov}(x, y) = 20$$

$$\text{Standard Deviation}(x) = \sqrt{\text{Variance}} = \sqrt{9} = 3$$

$$\text{Standard Deviation}(y) = \sqrt{\text{Variance}} = \sqrt{49} = 7$$

$$r = \frac{\text{cov}(x, y)}{(\sigma_x \cdot \sigma_y)} = \frac{20}{21} = 0.9523$$

Illustration 2

$\text{Cov}(x, y) = -100$, $V(x) = 400$ and $\text{SD}(y) = 5$ find r .

Solution

$$\text{Cov}(x, y) = -100, V(x) = 400, \sigma_y = 5$$

$$\begin{aligned} \text{Standard Deviation}(x) &= \sqrt{\text{Variance}} \\ &= \sqrt{400} = 20 \end{aligned}$$

$$r = \frac{\text{cov}(x, y)}{(\sigma_x \cdot \sigma_y)} = \frac{-100}{(20 \times 5)} = \frac{-100}{100} = -1$$

Illustration 3

Calculate Pearson's coefficient of correlation from the following data.

x	6	8	9	14	17	28	24
y	10	12	15	15	18	25	22

Solution

Calculation of Karl Pearson's co-efficient of correlation

x	y	xy	x ²	y ²
6	10	60	36	100
8	12	96	64	144
9	15	135	81	225
14	15	210	196	225
17	18	306	289	324
28	25	700	784	625
24	22	528	576	484
31	26	806	961	676
$\Sigma x = 137$	$\Sigma y = 143$	$\Sigma xy = 2841$	$\Sigma x^2 = 2987$	$\Sigma y^2 = 2803$

$\therefore$ There is a high degree of positive correlation.

$$\begin{aligned}
 r &= \frac{N \Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \times \sqrt{N \Sigma y^2 - (\Sigma y)^2}} \\
 &= \frac{8(2841) - (137)(143)}{\sqrt{8(2987) - (137)^2} \times \sqrt{8(2803) - (143)^2}} \\
 &= \frac{22728 - 19591}{\sqrt{23896 - 18769} \times \sqrt{22424 - 20449}} \\
 &= \frac{3137}{\sqrt{5127} \times \sqrt{1975}} = \frac{3137}{71.60 \times 44.44} \\
 &= \frac{3137}{3181.90} \Rightarrow r = 0.9858
 \end{aligned}$$

Illustration 4

Calculate Pearson's coefficient of correlation from the following data

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

Solution

x	y	x ²	y ²	xy
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\Sigma x = 70$	$\Sigma y = 63$	$\Sigma x^2 = 728$	$\Sigma y^2 = 651$	$\Sigma xy = 676$

$\therefore$ There is a high degree of positive correlation.

$$\begin{aligned}
 r &= \frac{N \Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \times \sqrt{N \Sigma y^2 - (\Sigma y)^2}} \\
 &= \frac{7(676) - (70)(63)}{\sqrt{7(728) - (70)^2} \times \sqrt{7(651) - (63)^2}} \\
 &= \frac{4732 - 4410}{\sqrt{5096 - 4900} \times \sqrt{4557 - 3969}} \\
 &= \frac{322}{\sqrt{196} \times \sqrt{588}} = \frac{322}{14 \times 24.24} = \frac{322}{339.36} \\
 r &= 0.9488
 \end{aligned}$$

Illustration 5

Calculate Karl Pearson's coefficient of correlation from the data given below

x	15	22	12	20	25
y	22	28	20	25	30

Solution

x	y	x^2	y^2	xy
15	22	225	484	330
22	28	484	784	616
12	20	144	400	240
20	25	400	625	500
25	30	625	900	750
$\Sigma x = 94$	$\Sigma y = 125$	$\Sigma x^2 = 1878$	$\Sigma y^2 = 3193$	$\Sigma xy = 2436$

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}}$$

$$= \frac{2436}{\sqrt{1878 \times 3193}} = \frac{2436}{2448.76} = 0.9949$$

(OR)

$$r = \frac{N \Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \times \sqrt{N \Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{5(2436) - (94)(125)}{\sqrt{5(1878) - (94)^2} \times \sqrt{5(3193) - (125)^2}}$$

$$= \frac{12180 - 11750}{\sqrt{9390 - 8836} \times \sqrt{15965 - 15625}}$$

$$r = \frac{430}{\sqrt{554} \times \sqrt{340}} = \frac{430}{23.53 \times 18.43} = \frac{430}{433.65} = 0.99$$

∴ There is a high degree of positive correlation.

Illustration 6

Find Correlation Coefficient for the given value under Karl Pearson's Method.

X	1	2	3	4	5
Y	4	5	6	7	8

Solution

X	Y	$x = X - \bar{X}$	$y = Y - \bar{y}$	xy	x^2	y^2
1	4	1 - 3 = -2	4 - 6 = -2	4	4	4
2	5	2 - 3 = -1	5 - 6 = -1	1	1	1
3	6	3 - 3 = 0	6 - 6 = 0	0	0	0
4	7	4 - 3 = 1	7 - 6 = 1	1	1	1
5	8	5 - 3 = 2	8 - 6 = 2	4	4	4
$\Sigma X = 15$	$\Sigma Y = 30$			$\Sigma xy = 10$	$\Sigma x^2 = 10$	$\Sigma y^2 = 10$

$$a. \bar{x} = \frac{\Sigma X}{n} = \frac{15}{5} = 3$$

$$b. \bar{y} = \frac{\Sigma Y}{n} = \frac{30}{5} = 6$$

$$c. r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{10}{\sqrt{10 \times 10}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

$$r = 1$$

∴ There is a Perfect Positive Correlation.

Illustration 7

Calculate Correlation Coefficient using Karl Pearson's Method

X	1	2	3	4	5
Y	10	9	8	7	6

Solution

Let A be x and B be y

X	Y	$x = X - \bar{x}$	$y = Y - \bar{y}$	xy	x^2	y^2
1	10	$1 - 3 = -2$	$10 - 8 = 2$	-4	4	4
2	9	$2 - 3 = -1$	$9 - 8 = 1$	-1	1	1
3	8	$3 - 3 = 0$	$8 - 8 = 0$	0	0	0
4	7	$4 - 3 = 1$	$7 - 8 = -1$	-1	1	1
5	6	$5 - 3 = 2$	$6 - 8 = -2$	-4	4	4
$\Sigma X = 15$	$\Sigma Y = 40$			$\Sigma xy = -10$	$\Sigma x^2 = 10$	$\Sigma y^2 = 10$

$$a. \bar{x} = \frac{\Sigma X}{n} = \frac{15}{5} = 3$$

$$b. \bar{y} = \frac{\Sigma Y}{n} = \frac{40}{5} = 8$$

$$c. r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{-10}{\sqrt{10 \times 10}} = \frac{-10}{\sqrt{100}} = \frac{-10}{10} = -1$$

$$r = -1$$

∴ There is a Perfect Negative Correlation.

Illustration 8

From the following table find out correlation between age and reading habits of students.

Age	15	16	17	18	19	20
No. of Students	250	200	150	120	100	80
Regular Readers	200	150	90	48	30	12

Solution

$$\text{Reading Habits} = \frac{\text{Regular Readers}}{\text{No. of Students}} \times 100$$

Note: Correlation between Age(x) and Reading Habits (y).

Age (X)	Reading habits(Y)	$x = X - \bar{x}$	$y = Y - \bar{y}$	xy	x^2	y^2
15	$\frac{200}{250} \times 100 = 80$	$15 - 17.5 = -2.5$	$80 - 50 = 30$	-75	6.25	900
16	$\frac{150}{200} \times 100 = 75$	$16 - 17.5 = -1.5$	$75 - 50 = 25$	-37.5	2.25	625
17	$\frac{90}{150} \times 100 = 60$	$17 - 17.5 = -0.5$	$60 - 50 = 10$	-5	0.25	100
18	$\frac{48}{120} \times 100 = 40$	$18 - 17.5 = 0.5$	$40 - 50 = -10$	-5	0.25	100
19	$\frac{30}{100} \times 100 = 30$	$19 - 17.5 = 1.5$	$30 - 50 = -20$	-30	2.25	400
20	$\frac{12}{80} \times 100 = 15$	$20 - 17.5 = 2.5$	$15 - 50 = -35$	-87.5	6.25	1225
$\Sigma X = 105$	$\Sigma Y = 300$			$\Sigma xy = -240$	$\Sigma x^2 = 17.5$	$\Sigma y^2 = 3350$

$$a. \bar{x} = \frac{\Sigma X}{n} = \frac{105}{6} = 17.5$$

$$c. \bar{y} = \frac{\Sigma Y}{n} = \frac{300}{6} = 50$$

$$b. r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{-240}{\sqrt{17.5 \times 3350}} = \frac{-240}{\sqrt{58625}} = \frac{-240}{242.13} = -0.99$$

$$r = -0.99$$

∴ There is a high degree of negative correlation.

Interpretation: The value of the correlation is highly negative, which means as age increases, the number of regular readers decreases. So, younger students are more regular in reading, while older students tend to read less regularly.

Illustration 9

From the following data find coefficient correlation by Karl Pearson's method between density of population and death rate.

Density of Population	200	500	400	700	600	300
Population (in 1000)	30	90	40	42	72	24
No. of death	300	1440	560	840	1224	312

Solution

Let us calculate Death Rate as under

$$\text{Death Rate} = \frac{\text{No. of death}}{\text{Population}}$$

X	Y	$x = X - \bar{x}$	$y = Y - \bar{y}$	xy	x^2	y^2
200	$\frac{300}{30} = 10$	$200 - 450 = -250$	$10 - 15 = -5$	1250	62500	25
500	$\frac{1440}{90} = 16$	$500 - 450 = 50$	$16 - 15 = 1$	50	2500	1
400	$\frac{560}{40} = 14$	$400 - 450 = -50$	$14 - 15 = -1$	50	2500	1
700	$\frac{840}{42} = 20$	$700 - 450 = 250$	$20 - 15 = 5$	1250	62500	25
600	$\frac{1224}{72} = 17$	$600 - 450 = 150$	$17 - 15 = 2$	300	22500	4
300	$\frac{312}{24} = 13$	$300 - 450 = -150$	$13 - 15 = -2$	300	22500	4
$\Sigma X = 2700$	$\Sigma Y = 90$			$\Sigma xy = 3200$	$\Sigma x^2 = 175000$	$\Sigma y^2 = 60$

a. $\bar{x} = \frac{\Sigma X}{n} = \frac{2700}{6} = 450$

b. $\bar{y} = \frac{\Sigma Y}{n} = \frac{90}{6} = 15$

c. $r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{3200}{\sqrt{175000 \times 60}} = \frac{3200}{\sqrt{10500000}} = \frac{3200}{3240.37} = 0.99$

$r = 0.99$

$\therefore$ There is a high degree of positive correlation.

Interpretation: The value of the correlation is highly positive, which means as the density population increases, the death rate also increases. This shows that areas with more people per space tend to have a higher number of deaths, possibly due to overcrowding, poor living conditions or lack of healthcare facilities.

Illustration 10

Following are the results of BBA Examination. Calculate Coefficient correlation between age and success in examination.

	20-21	21-22	22-23	23-24	24-25	25-26
Age	130	100	160	110	80	25
Candidates appeared	78	55	80	49.5	32	5
Success						

Solution

Mid values of Age group is taken as (x) and Success (y) is calculated as under

$$\text{Success in examination} = \frac{\text{Success}}{\text{Candidates appeared}} \times 100$$

Age (X)	Success (Y)	$x = X - \bar{x}$	$y = Y - \bar{y}$	xy	x^2	y^2
20.5	$\frac{78}{130} \times 100 = 60$	20.5 - 23 = -2.5	60 - 45 = 15	-37.5	6.25	225
21.5	$\frac{55}{100} \times 100 = 55$	21.5 - 23 = -1.5	55 - 45 = 10	-15	2.25	100
22.5	$\frac{80}{160} \times 100 = 50$	22.5 - 23 = -0.5	50 - 45 = 5	-2.5	0.25	25
23.5	$\frac{49.5}{110} \times 100 = 45$	23.5 - 23 = 0.5	45 - 45 = 0	0	0.25	0
24.5	$\frac{32}{80} \times 100 = 40$	24.5 - 23 = 1.5	40 - 45 = -5	-7.5	2.25	25
25.5	$\frac{5}{25} \times 100 = 20$	25.5 - 23 = 2.5	20 - 45 = -25	-62.5	6.25	625
$\Sigma X = 138$	$\Sigma Y = 270$			$\Sigma xy = -125$	$\Sigma x^2 = 17.50$	$\Sigma y^2 = 1000$

$$a. \bar{x} = \frac{\Sigma X}{n} = \frac{138}{6} = 23.$$

$$b. \bar{y} = \frac{\Sigma Y}{n} = \frac{270}{6} = 45.$$

$$c. r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{-125}{\sqrt{17.50 \times 1000}} = \frac{-125}{\sqrt{17500}} = \frac{-125}{132.29} = -0.94.$$

$$r = -0.94$$

There is a high degree of negative correlation.

Interpretation: The value of the correlation is highly negative, which means as age increases, the success rate in examination decreases. This shows that younger candidates are more likely to succeed in the BBA examination, while older candidates tend to have lower success rates.

Illustration 11

Find co-efficient of correlation between average sales and average profit per store.

No. of stores	15	20	25	30	25
Total Sales	9000	6000	10000	4500	4500
Total Profit	1500	4000	7500	1500	1500

Solution

$$X = \text{Average sales} = \frac{\text{Total sales}}{\text{No. of Stores}}$$

$$Y = \text{average Profit} = \frac{\text{Total Profit}}{\text{No. of Stores}}$$

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	xy	x^2	y^2
$\frac{9000}{15} = 600$	$\frac{1500}{15} = 100$	$600 - 326 = 274$	$100 - 142 = -42$	-11508	75076	1764
$\frac{6000}{20} = 300$	$\frac{4000}{20} = 200$	$300 - 326 = -26$	$200 - 142 = 58$	-1508	676	3364
$\frac{10000}{25} = 400$	$\frac{7500}{25} = 300$	$400 - 326 = 74$	$300 - 142 = 158$	11692	5476	24964
$\frac{4500}{30} = 150$	$\frac{1500}{30} = 50$	$150 - 326 = -176$	$50 - 142 = -92$	16192	30976	8464
$\frac{4500}{25} = 180$	$\frac{1500}{25} = 60$	$180 - 326 = -146$	$60 - 142 = -82$	11972	21316	6724
$\Sigma X = 1630$	$\Sigma Y = 710$			$\Sigma xy = -26840$	$\Sigma x^2 = 133520$	$\Sigma y^2 = 45280$

$$a. \bar{x} = \frac{\Sigma X}{n} = \frac{1630}{5} = 326$$

$$b. \bar{y} = \frac{\Sigma Y}{n} = \frac{710}{5} = 142$$

$$c. r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}} = \frac{-26840}{\sqrt{133520 \times 45280}} = \frac{-26840}{\sqrt{6045785600}} = \frac{-26840}{77754.65} = -0.35$$

$$r = -0.35$$

$\therefore$ There is a low degree of negative correlation.

Interpretation: The calculated correlation shows a low degree of negative correlation between average sales and average profit per store. This means that as average sales slightly increase, average profit does not increase in the same way and in some cases may even decrease. So, the relationship between sales and profit per store is weak and slightly inverse.

Illustration 12

Calculate Karl Pearson's co-efficient of correlation

Sum of deviation of $x = 5$ & $y = 4$

Sum of squares of the deviation of $x = 40$

Sum of squares of the deviation of $y = 50$

Sum of product of the deviation of x & $y = 32$

No. of observations = 10

Solution

$$\Sigma x = 5, \Sigma x^2 = 40, \Sigma xy = 32$$

$$\Sigma y = 4, \Sigma y^2 = 50, N = 10$$

$$\begin{aligned} r &= \frac{N\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N\Sigma x^2 - (\Sigma x)^2} \times \sqrt{N\Sigma y^2 - (\Sigma y)^2}} \\ &= \frac{10(32) - (5)(4)}{\sqrt{10(40) - (5)^2} \times \sqrt{10(50) - (4)^2}} \\ &= \frac{320 - 20}{\sqrt{400 - 25} \times \sqrt{500 - 16}} = \frac{300}{\sqrt{375} \cdot \sqrt{484}} \\ &= \frac{300}{19.36 \times 22} = \frac{300}{425.92} \end{aligned}$$

$$r = 0.7043$$

There is a moderate degree of positive correlation.

Illustration 13

$\Sigma x = 118$; $\Sigma y = 93$; $\Sigma x^2 = 556$; $\Sigma y^2 = 309$; $\Sigma xy = 368$; $n = 30$; find r .

Solution

$$\begin{aligned} r &= \frac{N\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N\Sigma x^2 - (\Sigma x)^2} \times \sqrt{N\Sigma y^2 - (\Sigma y)^2}} = \frac{30(368) - (118)(93)}{\sqrt{30(556) - (118)^2} \cdot \sqrt{30(309) - (93)^2}} \\ &= \frac{11040 - 10974}{\sqrt{16680 - 13924} \cdot \sqrt{9270 - 8649}} \\ &= \frac{66}{\sqrt{2756} \cdot \sqrt{621}} = \frac{66}{1308.3} = 0.0504 \end{aligned}$$

There is a low degree of positive correlation.

Illustration 1

If $r = 0.98$, $n = 5$, Examine whether r is significant or not.

Solution

$$\begin{aligned} P.E &= 0.6745 \left[\frac{1-r^2}{\sqrt{n}} \right] = 0.6745 \left[\frac{1-(0.98)^2}{\sqrt{5}} \right] = 0.6745 \left[\frac{1-0.9604}{2.23} \right] \\ &= 0.6745 \left[\frac{0.0396}{2.23} \right] = 0.6745(0.0177) = \mathbf{0.011} \end{aligned}$$

Here, $|r| > 6PE \Rightarrow 0.98 > 0.066$
 $6PE = 6(0.011) = 0.066$

$\therefore r$ is significant

Limits are

1. $r - PE = 0.98 - 0.011 = 0.969$
2. $r + PE = 0.98 + 0.011 = 0.991$

Interpretation : Since $r = 0.98$ is much greater than $6 \times PE = 0.066$, the correlation is said to be definitely significant. This means there is a very strong and meaningful relationship between the two variables, and the result is not due to chance.

The calculated limits 0.969 to 0.991 show the range in which the true population correlation is expected to lie. As both values are very close to +1, the correlation is almost perfect positive.

Illustration 2

If $r = -1$, $n = 4$, find whether r is significant or not.

Solution

a. $PE(r) = 0.6745 \times \frac{1-r^2}{\sqrt{n}} = 0.6745 \times \frac{1-(-1)^2}{\sqrt{4}} = 0.6745 \times \frac{1-(1)}{2} = 0.6745 \times \frac{0}{2} = 0.6745 \times 0 = 0$

b. $6PE = 6 \times 0 = 0$

c. Compare r and $6PE$: Since $|r| > 6PE$, the correlation is significant.

Interpretation: Even though r is negative, its absolute value (1) is much greater than $6PE$ (0). Hence the correlation is perfectly negative and statistically significant. There is a strong and definite relationship between the two variables.

Illustration 3

Calculate the probable error

X	45	70	65	30	90	40	50	75	85
Y	35	90	70	40	95	40	60	80	80

Solution						
X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	y^2	xy
45	35	-16	-29	256	841	464
70	90	9	26	81	676	234
65	70	4	6	16	36	24
30	40	-31	-24	961	576	744
90	95	29	31	841	961	899
40	40	-21	-24	441	576	504
40	60	-11	-4	121	16	44
50	80	14	16	196	256	224
75	80	24	16	576	256	384
85	50	-1	-14	1	196	14
60						
$\Sigma X = 610$	$\Sigma Y = 640$	$\Sigma x = 0$	$\Sigma y = 0$	$\Sigma x^2 = 3490$	$\Sigma y^2 = 4390$	$\Sigma xy = 3535$

$$\bar{x} = \frac{\Sigma X}{n} = \frac{610}{10} = 61; \quad \bar{y} = \frac{\Sigma Y}{n} = \frac{640}{10} = 64$$

$$r = \frac{N\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N\Sigma x^2 - (\Sigma x)^2} \sqrt{N\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{10(3535) - (0)(0)}{\sqrt{10(3490) - (0)^2} \sqrt{10(4390) - (0)^2}} = \frac{35350}{\sqrt{34900} \sqrt{43900}}$$

$$= \frac{35350}{(186.81)(209.52)} = \frac{35350}{39140.43} = 0.9031$$

$$\text{Probable Error} = 0.6745 \left(\frac{1-r^2}{\sqrt{n}} \right) = 0.6745 \left(\frac{1-(0.9031)^2}{\sqrt{10}} \right)$$

$$= 0.6745 \left(\frac{1-0.8155}{3.1622} \right) = 0.6745 \left(\frac{0.1845}{3.1622} \right)$$

$$= 0.6745 (0.0583) = 0.0393$$

$$|r| > 6 \text{ PE} \quad \Rightarrow \quad 0.9031 > 0.2358; \quad 6 \text{ PE} = 6 \times 0.0393 = 0.2358$$

$\therefore$ r is significant.

The limits are :

$$r - \text{PE} \quad \Rightarrow \quad 0.9031 - 0.0393 = 0.8638$$

$$r + \text{PE} \quad \Rightarrow \quad 0.9031 + 0.0393 = 0.9424$$

Interpretation : Since $|r| = 0.9031$ is greater than $6 \times \text{PE} = 0.2358$, the correlation is statistically significant. This means there is a strong and meaningful positive relationship between X and Y, and the true value of correlation lies between 0.8638 and 0.9424.

Illustration 4

Calculate 'r' and comment on the result of probable error

X	100	200	300	400	500	600
Y	120	130	140	150	160	170

Solution

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	xy	x^2	y^2
100	120	-250	-25	6250	62500	625
200	130	-150	-15	2250	22500	225
300	140	-50	-5	250	2500	25
400	150	50	5	250	2500	25
500	160	150	15	2250	22500	225
600	170	250	25	6250	62500	625
$\Sigma X = 2100$	$\Sigma Y = 870$	0	0	$\Sigma xy = 17500$	$\Sigma x^2 = 175000$	$\Sigma y^2 = 1750$

$$\bar{x} = \frac{\Sigma X}{n} = \frac{2100}{6} = 350$$

$$\bar{y} = \frac{\Sigma Y}{n} = \frac{870}{6} = 145$$

$$\begin{aligned}
 r &= \frac{N \Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \sqrt{N \Sigma y^2 - (\Sigma y)^2}} \\
 &= \frac{105000}{\sqrt{1050000} \sqrt{10500}} \\
 &= \frac{105000}{(104989.73)} = 1.00
 \end{aligned}$$

$$\begin{aligned}
 \text{Probable Error} &= 0.6745 \left(\frac{1-r^2}{\sqrt{n}} \right) = 0.6745 \left(\frac{1-1^2}{\sqrt{6}} \right) \\
 &= 0.6745 \left(\frac{0}{2.44} \right) = 0.6745(0) = 0
 \end{aligned}$$

$$|r| > 6 \text{ PE} \quad \Rightarrow \quad 1 > 0; \quad 6 \text{ PE} = 6 \times 0 = 0$$

$\therefore$ **r is significant**

The limits are:

$$r - \text{PE} \quad \Rightarrow \quad 1 - 0 = 1$$

$$r + \text{PE} \quad \Rightarrow \quad 1 + 0 = 1$$

Interpretation: The value of $r = 1$, which indicates a perfect positive correlation between X and Y. Since $\text{PE} = 0$, we get $6 \times \text{PE} = 0$, and clearly, $|r| > 6 \text{ PE}$, so the correlation is statistically significant. This means there is a perfect and reliable linear relationship between X and Y. As the value of X increases, Y also increases at a constant rate without any deviation. The true correlation lies exactly at 1, showing a perfect match between the two variables.

4.5.3 Spearman's Rank Correlation Coefficient

In real-life situations, we often come across data that is not in the form of exact numbers but in the form of ranks or order of performance. For example:

- Judges rank participants in a competition.
- Students are given class positions based on overall performance.
- People are rated based on preference (like most liked to least liked).

In such cases, using Karl Pearson's correlation is not suitable because that method works best with actual numerical values and assumes a normal distribution. We need a method that can handle non-numerical, ordinal, or rank-based data, and that is where **Spearman's Rank Correlation** becomes useful.

What is Spearman's Rank Correlation?

Spearman's Rank Correlation Coefficient (denoted by r_s) is a method used to measure the strength and direction of the relationship between two variables when the data is in the form of ranks. It helps to find out whether the ranks of two variables move together (positive correlation) or move in opposite directions (negative correlation).

It is useful just like Karl Pearson's Correlation, but instead of using actual values (like marks, sales, height), this method uses ranks or positions.

This method is especially helpful when we are not working with exact values but want to see the pattern in rankings. It is another way to check correlation—just like Karl Pearson's—but designed for ordered data instead of raw numbers.

When to Use Spearman's Method?

1. When ranks are already given

Example: Rank in maths and rank in science for each student.

2. When ranks are not given, but values can be arranged in order

Example: Marks, sales, age, height – these can be converted into ranks from highest to lowest or vice versa.

3. When two or more people have the same rank (called tied ranks)

In this case, we assign them the average rank.

Why Use Spearman's Rank Correlation?

In many situations, it is difficult to measure things with exact numerical values, but it is possible to arrange them in order. For example, it may not be easy to assign exact marks for a student's creativity, a person's behavior, or the quality of a product. But we can rank them from best to worst or most to least. In such cases, Spearman's Rank Correlation is very useful because it works with ranks instead of actual values.

Spearman's Rank Correlation is Useful When:

- Data is qualitative or subjective, like preferences, opinions, or ratings.
- The data is not normally distributed, meaning it does not follow a bell-shaped curve. Pearson's method is not suitable.
- The relationship is not linear but still has an order or pattern.
- The focus is on positions or relative comparison instead of exact values.

Real-Life Situations Where it is Used

- **Student Rankings in Two Subjects** : Suppose students are ranked based on performance in English and Maths. Even if marks are not available, we can use ranks to check if performance in one subject is linked with good performance in the other.
- **Employee Performance and Training Results** : Employees may be ranked by their supervisor based on job performance and also based on their training test results. Spearman's correlation helps to see if those who perform well also score better in training.
- **Customer Satisfaction and Complaint Records** : Companies can be ranked by customer satisfaction and also by the number of complaints received. This method helps to understand if lower satisfaction is linked with higher complaints.

It is especially helpful in fields like education, psychology, market research, and social sciences, where ranking-based analysis is more meaningful than exact numbers.

Formula (When no Tied Ranks)

In the rank correlation coefficient method, the ranks are given to each individual on the basis of its quality or quantity, such as ranking starts from position 1st and goes till Nth position for the ranked last in the group.

Formula:
$$r_s = 1 - \frac{6\sum d^2}{n^3 - n}$$

Where,

r_s = Rank coefficient of correlation

d = difference between the ranks of each pair ($R_1 - R_2$)

d^2 = square of that difference

n = total number of observations or pairs

Where Actual Ranks are Given

An individual must follow the following steps to calculate the correlation coefficient:

- a. First, the difference between the ranks ($R_1 - R_2$) must be calculated, denoted by d .
- b. Then, square these differences to remove the negative sign and obtain its sum $\sum d^2$.
- c. Apply the formula as shown above.

Where Ranks are not Given

- a. Assign rank 1 to the highest value, rank 2 to the next highest, and so on.
- b. Apply the same ranking method to both variables. Then follow the usual steps.

When Ranks are Repeated

- Sometimes two or more values are the same, so they get same ranks. In this case, give them the average of the positions they occupy.
- **Example:** If two students get the same marks and are both in 2nd position, instead of ranks 2 and 3, both will get: $(2+3)/2 = 2.5$
- If tied ranks are present, the following corrected formula is used:

$$r_s = 1 - \frac{6[\sum d^2 + CF]}{n^3 - n} \quad CF = \frac{m^3 - m}{12} + \dots$$

Where, CF = correction factor and m stands for no. of items which have common ranks.

Note: If there are multiple groups with tied ranks, find CF for each group and add them.

Range and Interpretation of r_s Values

The value of Spearman's Rank Correlation Coefficient (r_s) always lies between -1 and +1.

1. Direction of the Relationship:

- If $r_s > 0 \rightarrow$ The correlation is positive
This means as one variable increases, the other also tends to increase.
The ranks move in the same direction.
- If $r_s < 0 \rightarrow$ The correlation is negative
This means as one variable increases, the other tends to decrease.
The ranks move in the opposite direction.
- If $r_s = 0 \rightarrow$ There is no relationship between the two sets of ranks.
The rankings do not follow any pattern.

2. Strength of the Relationship:

The numerical value of r_s (ignoring the sign) shows the strength of the correlation:

- A value close to +1 or -1 indicates a strong relationship
- A value close to 0 indicates a weak or no relationship

This tells how closely the two variables follow the same (or opposite) order.

Summary Table

r_s Value Range	Strength of Relationship	Direction	Interpretation
+1.00	Perfect	Positive (same direction)	Complete agreement in ranks. Both variables move in the exact same order.
+0.80 to +0.99	Very Strong	Positive	Ranks are highly similar. As one rank increases, the other also increases.
+0.60 to +0.79	Strong	Positive	A clear positive pattern exists in the rankings.

+0.40 to +0.59	Moderate	Positive	Some positive moderate agreement in ranking, though not perfect.
+0.20 to +0.39	Weak	Positive	Slight similarity in ranks; the relationship is weak.
+0.00 to +0.19	Very Weak or None	Positive	Very little or no ranking relationship.
0	None	No direction	No relationship at all between the sets of ranks.
-0.01 to -0.19	Very Weak or None	Negative (opposite)	Very small tendency for opposite ranking.
-0.20 to -0.39	Weak	Negative	A weak reverse relationship in ranking.
-0.40 to -0.59	Moderate	Negative	Moderate level of disagreement in ranking.
-0.60 to -0.79	Strong	Negative	Clear opposite trend in the rankings.
-0.80 to -0.99	Very Strong	Negative	High disagreement in ranks. As one increases, the other decreases.
-1.00	Perfect	Negative	Complete opposite ranking. Perfect reverse order between the two variables.

Problems: (When Ranks are Given)

Illustration 1

Two judges in beauty contest ranked the 10 entries as follows compute the ranks correlation coefficient and interpret the result.

X	1	2	3	4	5	6	7	8	9	10
Y	8	7	6	4	3	2	5	1	9	10

Solution

R_1	R_2	$d = R_1 - R_2$	d^2
1	8	-7	49
2	7	-5	25
3	6	-3	9
4	4	0	0
5	3	2	4
6	2	4	16
7	5	2	4
8	1	7	49
9	9	0	0
10	10	0	0
			$\Sigma d^2 = 156$

$$r_s = 1 - \frac{6 \Sigma d^2}{n^3 - n} = 1 - \frac{6(156)}{(10)^3 - 10}$$

$$= 1 - \frac{936}{1000 - 10} = 1 - \frac{936}{990} = 1 - 0.9454 = 0.0546$$

Interpretation:

The calculated value of $r_s = 0.0546$, which is positive, indicates the direction of the relationship is positive. This means that as one judge gives a higher rank, the other tends to give the same. However, since the value is very close to 0, the strength of the relationship is very weak or almost none. So, while the judges may slightly agree in some places, overall their rankings are quite different, and there is no strong agreement between them.

Illustration 2

In a competition

Judge 1

Judge 2

Solution

R_1

3

4

1

5

2

6

10

9

8

7

Interpretation

The value of r_s is positive, since it is very close to 1, there is a strong agreement between the two judges.

Illustration 3

There are three judges and 10 entries. The following table shows the ranks assigned by each judge.

Judge A

Judge B

Judge C

Solution

R_1	R_2
1	3
6	5
5	8
10	4
3	7
2	10

Illustration 2

In a competition two judges ranks for 10 participants as follows:

	3	4	1	5	2	6	10	9	8	7
Judge 1										
Judge 2	6	10	3	9	5	2	8	7	4	1

Solution

R_1	R_2	$d = R_1 - R_2$	d^2
3	6	-3	9
4	10	-6	36
1	3	-2	04
5	9	-4	16
2	5	-3	09
6	2	4	16
10	8	+2	04
9	7	2	04
8	4	4	16
7	1	6	36
			$\Sigma d^2 = 150$

$$r_s = 1 - \frac{6\Sigma d^2}{n^3 - n} = 1 - \frac{6(150)}{(10)^3 - 10}$$

$$= 1 - \frac{900}{1000 - 10} = 1 - \frac{900}{990} = 1 - 0.9090$$

$$r_s = 0.091$$

Interpretation:

The value of $r_s = 0.091$ is positive and it shows a slight agreement in ranking direction. However, since it is very close to 0, the strength of the relationship is very weak or none. So, there is almost no agreement between Judge 1 and Judge 2 in their rankings.

Illustration 3

There are 10 competitors in a music competition ranked by three judges A, B & C in the following order.

Judge A	1	6	5	10	3	2	4	9	7	8
Judge B	3	5	8	4	7	10	2	1	6	9
Judge C	6	4	9	8	1	2	3	10	5	7

Solution

R_1	R_2	R_3	$d_1 = R_1 - R_2$	d_1^2	$d_2 = R_2 - R_3$	d_2^2	$d_3 = R_1 - R_3$	d_3^2
1	3	6	-2	4	-3	9	-5	25
6	5	4	1	1	1	1	2	4
5	8	9	-3	9	-1	1	-4	16
10	4	8	6	36	-4	16	2	4
3	7	1	-4	16	6	36	2	4
2	10	2	-8	64	8	64	0	0

4	2	3	2	04	-1	01	
9	1	10	8	64	-9	81	1
7	6	5	1	01	1	01	-1
8	9	7	-1	01	2	04	2
				$\Sigma d_1^2 = 200$		$\Sigma d_2^2 = 214$	1

$$r_s = 1 - \frac{6\Sigma d_1^2}{n^3 - n} = 1 - \frac{6(200)}{(10)^3 - 10}$$

$$= 1 - \frac{1200}{1000 - 10}$$

$$= 1 - \frac{1200}{990} = 1 - 1.2121$$

$$r_s = -0.2121$$

$$r_s = 1 - \frac{6\Sigma d_2^2}{n^3 - n} = 1 - \frac{6(214)}{(10)^3 - 10}$$

$$= 1 - \frac{1284}{1000 - 10}$$

$$= 1 - \frac{1284}{990} = 1 - 1.2969$$

$$r_s = -0.2969$$

$$r_s = 1 - \frac{6\Sigma d_3^2}{n^3 - n} = 1 - \frac{6(360)}{(10)^3 - 10}$$

$$= 1 - \frac{360}{1000 - 10}$$

$$= 1 - \frac{360}{990} = 1 - 0.3636$$

$$r_s = 0.6364$$

Interpretation:

- **rs between Judge A and Judge B = -0.2121**

This is a negative and weak correlation, meaning there is no proper agreement between Judge A and Judge B.

- **rs between Judge B and Judge C = -0.2969**

This is also negative and weak, showing no agreement between Judge B and Judge C.

- **rs between Judge A and Judge C = 0.6364**

This is a strong positive correlation, which means there is a strong level of agreement between Judge A and Judge C in ranking the participants.

Problems on : (When Actual Ranks are not Given)

Illustration 1

Calculate Spearman's co-efficient of Rank correlation from the following:

X	48	60	72	62	56	40	39	52
Y	63	79	66	71	39	55	61	33

Solution

To assign ranks, the values in each series (X and Y) are first arranged in descending order. Higher values are given higher ranks (Rank 1 for the highest value, Rank 2 for the next highest, and so on). For the X values, 72 is the highest, so it is given Rank 1, followed by 62 as Rank 2, and so on. The smallest value 30 is given Rank 9. The same process is followed for the Y values: 79 is the highest, so it gets Rank 1, followed by 71 as Rank 2, and so on. If any values are equal (tied), they are given the average of their positions, but in this case, all values are different, so unique ranks are assigned directly.

Assign ranks to the given data

X	Y	R_1	R_2	$d = R_1 - R_2$	d^2
48	63	6	4	2	4
60	79	3	1	2	4
72	66	1	3	-2	4
62	71	2	2	0	0
56	39	4	7	-3	9
40	55	7	6	1	1
39	61	8	5	3	9
52	33	5	8	-3	9
30	32	9	9	0	0
					$\Sigma d^2 = 40$

$$r_s = 1 - \frac{6\Sigma d^2}{n^3 - n}$$

$$= 1 - \frac{6(40)}{(9)^3 - 9} \Rightarrow 1 - \frac{240}{729 - 9} \Rightarrow 1 - \frac{240}{720} = 1 - 0.3333$$

$$r_s = 0.6666$$

Interpretation: The calculated value of Spearman's rank correlation (r_s) = 0.6666, which is positive. This shows a strong positive relationship between the two sets of ranks. It means that as values of X increase, the values of Y also tend to increase, and there is a good level of agreement between the two rankings.

Illustration 2

Calculate Rank correlation Co-efficient between advertisement and sales from the following data.

Advertisement	40	66	63	91	83	76	26	99	37	79
Sales	48	54	59	87	63	69	61	92	52	85

Solution

Assign ranks to the given data

Advertisement	Sales	R_1	R_2	$d = R_1 - R_2$	d^2
40	48	8	10	-2	4
66	54	6	8	-2	4
63	59	7	7	0	0

91	87	2	2	0	0
83	63	3	5	-2	4
76	69	5	4	1	1
26	61	10	5	4	16
99	92	1	1	0	0
37	52	9	9	0	0
79	85	4	3	1	1
					$\Sigma d^2 = 18$

$$r_s = 1 - \frac{6\Sigma d^2}{n^3 - n} = 1 - \frac{6(30)}{(10)^3 - 10} = 1 - \frac{180}{1000 - 10}$$

$$= 1 - \frac{180}{990} = 1 - 0.1818$$

$$r_s = 0.8182$$

Interpretation: The value of $r_s = 0.8182$, which is positive and very strong. This shows that there is a strong positive relationship between advertisement spending and sales. It means that when advertisement increases, sales also tend to increase, and the ranking pattern is fairly consistent.

Illustration 3

Calculate Spearman's co-efficient of correlation from the following information.

X	53	98	95	81	75	61	59	55
Y	47	25	32	37	30	40	39	45

Solution

Assign ranks to the given data

X	Y	R_1	R_2	$d = R_1 - R_2$	d^2
53	47	8	1	7	49
98	25	1	8	-7	49
95	32	2	6	-4	16
81	37	3	5	-2	4
75	30	4	7	-3	9
61	40	5	3	2	4
59	39	6	4	2	4
55	45	7	2	5	25
					$\Sigma d^2 = 160$

$$r_s = 1 - \frac{6\Sigma d^2}{n^3 - n}$$

$$= 1 - \frac{6(160)}{8^3 - 8} = 1 - \frac{960}{512 - 8}$$

$$= 1 - \frac{960}{504} = 1 - 1.9047$$

$$r_s = -0.9047$$

Interpretation : The value of $r_s = -0.9048$, which is a very strong negative correlation. This means there is a very strong disagreement between the ranks of X and Y. As values of X increase, values of Y tend to decrease, showing that both sets of rankings move in opposite directions.

Problems on : When Ranks are Repeated

Illustration 1

Calculate co-efficient of Rank correlation from the following information.

Marks in Economics	26	37	39	23	51	71	37	96
Marks in Accountancy	60	50	70	50	40	30	50	80

Solution

Assign ranks to the given data

Let X be economics & Y is accountancy

X	Y	R ₁	R ₂	d = R ₁ - R ₂	d ²
26	60	7	3	4	16
37	50	5.5	5	0.5	0.25
39	70	4	2	2	4
23	50	8	5	3	9
51	40	3	7	-4	16
71	30	2	8	-6	36
37	50	5.5	5	0.5	0.25
96	80	1	1	0	0
					$\Sigma d^2 = 81.5$

In X series 37 repeated 2 times

$$\boxed{37} = \frac{5+6}{2} = \frac{11}{2} = 5.5$$

In Y series 50 repeated 3 times

$$\boxed{50} = \frac{4+5+6}{3} = \frac{15}{3} = 5$$

$$\text{Correction factor} = \frac{m^3 - m}{12}$$

m = No. of times value repeated

$$\boxed{37} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5$$

$$\boxed{50} = \frac{m^3 - m}{12} = \frac{3^3 - 3}{12} = \frac{27 - 3}{12} = \frac{24}{12} = 2$$

$$CF = 0.5 + 2 = 2.5$$

$$r_s = 1 - \frac{6(\Sigma d^2 + CF)}{n^3 - n} = 1 - \frac{6(81.5 + 2.5)}{8^3 - 8}$$

$$= 1 - \frac{6(84)}{512 - 8} = 1 - \frac{504}{504} = 1 - 1$$

$$r_s = 0$$

Interpretation: The value of $r_s = 0$, which shows no rank correlation between marks in Economics and Accountancy. This means there is no consistent pattern or relationship between how marks scored in the two subjects. The rankings do not move together in any direction. In other words, a person who scores well in Economics does not necessarily score well in Accountancy, and vice versa.

Illustration 2

From the following data, calculate the co-efficient of rank correlation between x & y.

X	43	66	60	75	54	48	54	60	25
Y	60	45	80	35	45	68	85	70	65

Solution

Assign ranks to the given data

X	Y	R_1	R_2	$d = R_1 - R_2$	d^2
43	60	8	6	2	4
66	45	2	7.5	-5.5	30.25
60	80	3.5	2	1.5	2.25
75	35	1	9.5	-8.5	72.25
54	45	5.5	7.5	-2	4
48	68	7	4	3	9
54	85	5.5	1	4.5	20.25
60	70	3.5	3	0.5	0.25
25	65	10	5	5	25
36	35	9	9.5	-0.5	0.25
					$\Sigma d^2 = 167.5$

In X series 60 repeated 2 times :

$$\boxed{60} = \frac{3+4}{2} = \frac{7}{2} = 3.5$$

In X series 54 repeated 2 times

$$\boxed{54} = \frac{5+6}{2} = \frac{11}{2} = 5.5$$

In Y series 45 repeated 2 times

$$\boxed{45} = \frac{7+8}{2} = \frac{15}{2} = 7.5$$

In Y series 35 repeated 2 times

$$\boxed{35} = \frac{9+10}{2} = \frac{19}{2} = 9.5$$

$$\text{Correction factor} = \frac{m^3 - m}{12}$$

m = No. of times value repeated

$$\boxed{60} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5 \quad \boxed{54} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5$$

$$\boxed{45} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5 \quad \boxed{35} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5$$

$$CF = 0.5 + 0.5 + 0.5 + 0.5 = 2$$

$$r_s = 1 - \frac{6[\Sigma d^2 + CF]}{n^3 - n} = 1 - \frac{6(167.5 + 2)}{10^3 - 10}$$

$$= 1 - \frac{6(169.5)}{1000 - 10} = 1 - \frac{1017}{990} = 1 - 1.0273$$

$$r_s = -0.0273$$

Interpretation: The value of $r_s = -0.0273$, which is very close to zero and negative. This means there is almost no correlation between variables X and Y. In simple words, a high score in X does not mean a high or low score in Y. The rankings do not follow any consistent pattern between the two variables.

Illustration 3

Calculate coefficient of rank correlation from the following information.

X	48	33	40	9	16	16	65	24	16	57
Y	13	31	31	6	15	4	20	9	06	19

Solution

Assign ranks to the given data

X	Y	R_1	R_2	d	d^2
48	13	3	6	-3	9
33	31	5	1.5	3.5	12.25
40	31	4	1.5	2.5	6.25
9	6	10	8.5	1.5	2.25
16	15	8	5	3	9
16	4	8	10	-2	4
65	20	1	3	-2	4
24	9	6	7	-1	1
16	6	8	8.5	-0.5	0.25
57	19	2	4	-2	4
				$\Sigma d^2 = 52$	

In X series 16 repeated 3 times

$$\boxed{16} = \frac{7+8+9}{3} = 8$$

In Y series 31 repeated 2 times

$$\boxed{31} = \frac{1+2}{2} = 1.5$$

In Y series 6 repeated 2 times

$$\boxed{6} = \frac{8+9}{2} = 8.5$$

$$\text{Correction factor} = \frac{m^3 - m}{12}$$

m = No. of times value repeated

$$\boxed{16} = \frac{m^3 - m}{12} = \frac{3^3 - 3}{12} = \frac{27 - 3}{12} = \frac{24}{12} = 2$$

$$\boxed{31} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5$$

$$\boxed{6} = \frac{m^3 - m}{12} = \frac{2^3 - 2}{12} = \frac{8 - 2}{12} = \frac{6}{12} = 0.5$$

$$CF = 2 + 0.5 + 0.5 = 3$$

$$r_s = 1 - \frac{6(\sum d^2 + cf)}{n^3 - n} = 1 - \frac{6(52 + 3)}{(10)^3 - 10} = 1 - \frac{6(55)}{1000 - 10}$$

$$= 1 - \frac{330}{990} = 1 - 0.3333 = 0.6667$$

$$r_s = 0.6667$$

Interpretation: The value of $r_s = 0.6667$, which is positive and strong. This means there is an agreement or positive relationship between the two sets of rankings. So, as X values rise or fall, Y values tend to move in the same direction, though not perfectly.

4.6 Review Questions

Short Answer Type Questions

1. What is correlation?
2. Define coefficient of correlation.
3. What are the limits of r ?
4. What is positive correlation? Give one example.
5. What is negative correlation? Give one example.
6. What is linear correlation?
7. What is non-linear correlation?
8. What is simple correlation?
9. What is multiple correlation?
10. Write one importance of correlation.
11. State Karl Pearson's formula for correlation.
12. Give the formula for Spearman's Rank correlation co-efficient
13. What is a scatter diagram?
14. What does $r = 0$ indicate?
15. What does $r = +1$ indicate?
16. What does $r = -1$ indicate?
17. What is probable error?
18. State the formula for probable error of r .

19. What is SP?
20. Write the
21. When do
22. Distinguish
23. State the
24. Mention
25. If $\text{Var}(x)$
26. $\text{Cov}(x, y)$
27. If $\sum xdy$
28. If $r = 0$
29. If $r = 0$
30. If $b_{xy} =$
31. If $b_{xy} =$
32. $\sum d^2 =$
33. $\sum d^2 =$

Long Answer

1. Explain
2. Write
3. Explain
4. Explain
5. Explain
6. Explain
7. Explain
8. Explain
9. Explain
10. State
11. Explain
12. Explain
13. Distinguish
14. Explain
15. Write
16. In
17. C
18. .

19. What is Spearman's rank correlation?
20. Write the formula of Spearman's rank correlation.
21. When do you say Correlation is Positive? Give example.
22. Distinguish between Positive and Negative Correlation.
23. State the types of Correlation.
24. Mention the uses of Correlation?
25. If $\text{Var}(x) = 16$, $\text{Var}(y) = 25$ and $\text{Cov}(x, y) = 20$, find r .
26. $\text{Cov}(x, y) = -40$, $\sigma_x = 8$, $\sigma_y = 10$. Find correlation coefficient.
27. If $\sum dx dy = 120$, $\sum dx^2 = 240$, $\sum dy^2 = 360$, find r .
28. If $r = 0.7$ and $n = 25$, find probable error (PE).
29. If $r = 0.5$, $\text{PE} = 0.05$, check whether correlation is significant.
30. If $b_{xy} = 0.8$ and $b_{yx} = 0.6$, find r .
31. If $b_{xy} = 1.2$ and $b_{yx} = 0.9$, find r .
32. $\sum d^2 = 24$, $n = 6$. Find Spearman's rank correlation coefficient.
33. $\sum d^2 = 40$, $n = 8$. Find Spearman's rank correlation coefficient.

Long Answer Type Questions

1. Explain the meaning and definitions of correlation.
2. Write the importance and uses of correlation in data analysis.
3. Explain the types of correlation based on direction of change with examples.
4. Explain the types of correlation based on the form of relationship with examples.
5. Explain the types of correlation based on the number of variables studied.
6. Explain the concept of correlation coefficient and its interpretation.
7. Explain the degrees of correlation with suitable examples.
8. Explain the scatter diagram method of measuring correlation.
9. Explain Karl Pearson's coefficient of correlation with formula.
10. State the assumptions, merits, and demerits of Karl Pearson's correlation.
11. Explain the meaning and uses of probable error of correlation.
12. Explain Spearman's rank correlation method with formula and uses.
13. Differentiate between Karl Pearson's correlation and Spearman's rank correlation.
14. Explain the relationship between correlation and regression.
15. Write short notes on nonsense correlation.
16. In a bivariate data on x and y , $\text{Var}(x) = 16$, $\text{Var}(y) = 36$, $\text{Cov}(x, y) = 18$. Find r .
17. $\text{Cov}(x, y) = -75$, $\text{Var}(x) = 225$, $\sigma_y = 12$. Find r .
18. Calculate Pearson's coefficient of correlation from the following data:

x:	5	7	10	15	18	25	20	30
y:	12	14	16	18	20	28	24	32

19. Calculate Karl Pearson's coefficient of correlation from the following data:

x:	14	11	9	12	15	17	8
y:	10	7	6	9	11	13	4

20. Calculate Karl Pearson's coefficient of correlation from the data:

x:	18	20	15	22	28
y:	25	27	22	28	35

21. Find correlation coefficient by Karl Pearson's method:

x:	2	4	6	8	10
y:	12	14	16	18	20

22. Calculate correlation coefficient using Karl Pearson's method:

x:	1	2	3	4	5
y:	15	13	11	9	7

23. From the following table, find correlation between age and reading habits of students:

Age:	15	16	17	18	19	20
No. of Students:	240	210	160	130	90	70
Regular Readers:	190	160	100	60	40	20

24. From the following data, find coefficient of correlation between density of population and death rate:

Density:	250	400	600	500	700	300
Population (000s):	35	45	60	55	75	25
No. of Deaths:	350	600	1020	825	1260	275

25. Following are the results of BCA Examination. Calculate coefficient of correlation between age and success in examination:

Age:	20-21	21-22	22-23	23-24	24-25	25-26
Candidates Appeared:	140	120	150	100	90	30
Success:	84	60	75	45	36	6

26. Find coefficient of correlation between average sales and average profit per store:

No. of Stores:	10	20	25	30	15
Total Sales:	5000	7000	8000	6000	4000
Total Profit:	1500	3500	5000	2000	1200

27. $\Sigma x = 120$, $\Sigma y = 95$, $\Sigma x^2 = 600$, $\Sigma y^2 = 320$, $\Sigma xy = 400$, $n = 25$. Find r .

28. If $r = 0.92$, $n = 6$, test whether correlation is significant.
 29. If $r = -1$, $n = 5$, test whether correlation is significant.
 30. Calculate the probable error of r from the data:

x:	40	55	60	35	90	45	65	80	85	70
y:	30	60	65	40	100	50	70	75	85	55

31. Calculate r and comment on probable error from the following data:

x:	50	100	150	200	250	300
y:	60	75	90	105	120	135

32. Two judges ranked 8 candidates as follows. Calculate Spearman's rank correlation:

Judge 1:	1	3	2	4	5	6	7	8
Judge 2:	2	4	1	3	6	5	8	7

33. In a competition two judges ranked 6 students as follows. Find rank correlation:

Judge A:	3	4	1	5	2	6
Judge B:	2	6	3	4	5	1

34. Calculate Spearman's coefficient of correlation from the following data:

X:	45	70	65	30	90	40	50	75
Y:	35	90	70	40	95	35	55	80

35. Calculate rank correlation coefficient between advertisement and sales:

Advertisement:	30	60	50	90	70	80	40	100
Sales:	40	55	65	85	60	75	45	95

36. Find out Karl Pearson's Co-efficient of Correlation from the following data between Income and Weight.

Income in ₹	100	200	300	400	500	600
Weight in Kgs	120	130	140	150	160	170

37. If covariance between X and Y variables is 12.5 and the variance of X and Y are respectively 16.4 and 13.8, find the co-efficient of correlation between them.
 38. For a given distribution the value of correlation is 0.64 and its probable error is 0.13274. Find the number of items in series.
 39. Calculate Spearman's rank correlation co-efficient from the following data

X	60	15	20	28	12	40	80	20
Y	10	40	30	50	30	20	60	30

Unit - II

Chapter

5

REGRESSION ANALYSIS

5.1 Regression Analysis

5.1.1 Introduction

In the previous topic, we learned about correlation, which helps to understand whether two variables move together and in which direction. Now, we move one step further to study regression, which not only shows the relationship between variables but also helps in making predictions based on that relationship.

The concept of regression was first introduced by Sir Francis Galton in the late 19th century while studying the relationship between the heights of parents and their children. He observed that tall parents tend to have children who are slightly shorter than themselves, and short parents tend to have children who are slightly taller — a tendency he called "regression toward the mean."

Over time, regression evolved into a powerful statistical tool used in almost every field — economics, science, business, engineering, agriculture, and social sciences. The simplest form of regression is linear regression, which studies the relationship between two variables in the form of a straight line.

In business, several times it becomes necessary to have some forecast so that the management can take a decision regarding a product or a particular course of action. In order to make a forecast, one has to ascertain some relationship between two or more variables relevant to a particular situation. For example, a company is interested to know how far the demand for television will increase in the next five years, keeping in mind the growth of population in a certain town. Here, it clearly assumes that the increase in population will lead to an increased demand for television. Thus, to determine the nature and extent of relationship between these two variables becomes important for the company.

Example

In daily life and business, there is a need to predict or estimate something in the future. For example:

- A company may want to know how much sales will increase if they spend more on advertising.
- A school may want to know whether extra study hours help in improving exam marks.
- A shopkeeper may want to know if temperature affects the sale of cold drinks.

In all these cases, we try to understand the relationship between two things (called variables). This method of finding the relationship and using it to predict values is called Regression Analysis.

5.1.2 What is Regression and Regression Analysis?



Meaning: Regression and Regression Analysis

1. **Regression** is a method to find the average relationship between two or more variables. It is used to predict the value of one thing by using the known value of another.

Example: If a shopkeeper knows that every Rs.1,000 spent on advertising brings Rs.10,000 in sales, then he can use this relationship to predict how much to expect in sales if he spends Rs.2,000 on advertising.

So, regression helps in answering questions like:

- "What will happen to Y if X increases?"
- "How much increase in sales can I expect if I increase my advertisement?"

2. **Regression Analysis** is a statistical tool used to determine the probable change in one variable for a given amount of change in another. This means, the value of the unknown variable can be estimated from the known value of another variable. The regression analysis expresses the relationship between two or more variables in the form of an equation to estimate the value of a variable, based on the given value of another variable.



Definition: Regression and Regression Analysis

1. According to Blair, "Regression is the measure of the average relationship between two or more variable in terms of the original units of the data".
2. According to Wallis and Robert, "One of the most frequently used techniques in economics and business research, to find a relation between two or more variable that are related casually, is regression analysis".

The regression analysis is widely used in all the scientific disciplines. In economics, it plays a significant role in measuring or estimating the relationship among the economic variables. For example, the variables - price (x) and demand (y) are closely related to each other, so we can find out the probable value of x from the given value of y and similarly the probable value of y can be found out from the given value of x.

5.1.3 Dependent and Independent Variables

In simple regression, there are only two variables. One variable is used to predict or estimate the other. These variables are named based on how they behave in the relationship:

- a. **Dependent Variable** : The dependent variable is the one that is being predicted or estimated. It depends on another variable for its value.

Example: If we want to estimate sales based on advertisement cost, then sales is the dependent variable.

- b. **Independent Variable** : The independent variable is the one that is used to make the prediction. It is also called the explanatory or predictor variable.

Example: In the same case, advertisement cost is the independent variable, because we use it to estimate sales.

Example

When we are trying to predict the demand for television sets on the basis of population growth, we are using the demand for television sets as the dependent variable and the population growth as the independent or predictor variable.

Uses of Regression Analysis**5.1.1****1. To Predict the Value of one Variable from Another**

Regression helps to estimate the value of a dependent variable (like sales) based on a known independent variable (like advertisement cost).

Example : If a company knows that Rs.1,000 in advertising usually leads to Rs.10,000 in sales, they can predict future sales based on different ad budgets.

2. To Measure the Accuracy of Prediction

The standard error in regression shows how much error or uncertainty is involved in predictions made using the regression line.

Example : If a store uses past customer spending to predict next month's revenue, standard error tells how close the prediction is likely to be.

3. To Calculate Correlation from Regression

The regression coefficients can be used to calculate the correlation coefficient, which tells how strong the relationship is between variables.

Example : If marks and study hours are studied using regression, the coefficient also shows how strongly they are related.

4. To Study Cause-and-Effect Relationships

Regression is widely used in research to explore whether changes in one variable cause changes in another.

Example : A health researcher might use regression to find out if increased exercise leads to better heart health.

5. To Make Business Decisions

Businesses use regression to decide pricing, production, or budgeting by estimating how changes in cost or market trends will affect profits.

Example : A company may use regression to estimate how reducing the price of a product will impact sales volume.

6. To Forecast Future Trends

Regression is a tool for forecasting in economics, finance, and production planning.

Example : Economists use regression to predict future inflation based on factors like interest rates and supply chain costs.

7. To Identify Important Factors

In multiple regression, it helps to find out which variables have the most impact.

Example : In predicting house prices, regression can show whether location, size, or age of the house affects price the most.

8. To Support Data-Driven Planning

Regression analysis provides numerical support for making future plans based on past data.

Example: A school may plan extra classes if regression shows that more study hours lead to better final exam performance.

5.1.5 Types of Regression

Regression can be classified into different types depending on the number of variables, the nature of the relationship, and how many factors are considered in the analysis.

1. Simple and Multiple Regression

- **Simple Regression:** This involves only two variables – one independent variable (X) and one dependent variable (Y).

Example: Predicting sales based on advertising cost. (More ads → more sales)

- **Multiple Regression:** This involves more than two variables – one dependent variable and two or more independent variables.

Example: Predicting sales using advertising, price, and store location.

2. Linear and Non-Linear Regression

- **Linear Regression:** The relationship between the variables forms a straight line when plotted on a graph. The equation used is simple, and the change is in a constant ratio.

Example: As study hours increase, exam marks increase in a steady manner.

- **Non-Linear Regression:** The relationship is curved or follows a non-straight path. Here, the effect of change in X on Y is not consistent.

Example: Exercise improves health, but too much exercise can lead to exhaustion. The curve rises first, then drops.

3. Total and Partial Regression

- **Total Regression:** All the important variables that affect the outcome are considered.

Example: To predict crop yield, we consider rainfall, soil quality, fertilizer used, etc.

- **Partial Regression:** Only some key variables are used for prediction. Other variables are ignored if they are not relevant for the purpose.

Example: Predicting exam results using only study time, without considering health or class attendance.

5.1.6 Regression Lines

A regression line is a straight line that shows the relationship between two variables on a graph. It helps to predict the value of one variable when the value of the other variable is known.

When data points are plotted on a graph, they are usually scattered. The regression line is drawn such a way that it goes as close as possible to all the points. This line minimizes the total distance between the line and all the data points. That is why it is called the **line of best fit**.

Example

Suppose a company

- In January,
- In February,
- In March, the

These points are plotted. The middle of all the points will be above the line, showing the general trend: increasing.

Two Types of Regression

When we study the relationship between two variables, it is called bivariate regression. There are two types of regression lines:

1. **Regression Line of Y on X:** This line is called the regression line of Y on X.

Example: Predicting sales based on advertising cost.

2. **Regression Line of X on Y:** This line is called the regression line of X on Y.

Example: Predicting advertising cost based on sales.

Important Note:

- There are two types of regression lines.
- The algebraic equations for these lines are different.
- Each line is used for a different purpose.
- There will be a difference in the interpretation of the results.
- These equations are used to predict the value of one variable when the value of the other variable is known.

Why are There Two Types?

Because either variable can be the dependent variable.

- One line is used to predict the value of Y when X is known.
- The other line is used to predict the value of X when Y is known.

These lines are called regression lines.

Understanding the Regression Line

The below diagram shows the relationship between two variables — X and Y.

Example

Suppose a company is checking how advertising expenses (X) affect sales (Y). They collect data for a few months:

- In January, they spent Rs.5,000 and earned Rs.50,000
- In February, they spent Rs.10,000 and earned Rs.1,00,000
- In March, they spent Rs.7,000 and earned Rs.70,000

These points are plotted on a graph as dots. Now, a regression line is drawn in such a way that it goes through the middle of all these points. It does not pass through every point, but stays close to most of them. Some dots will be above the line, some below, but the line will be in the middle, balancing the distances. This line shows the general trend: more advertising $\rightarrow$ more sales.

Two Types of Regression Lines

When we study the relationship between two variables (like X = advertisement cost and Y = sales), it is called bivariate data. In such cases, we can use one variable to predict the other. So, there will be two regression lines depending on which variable we are using to predict the other.

1. **Regression Line of Y on X** : This line helps to predict the value of Y when X is known. This is called regression of Y on X.

Example: If we know the advertisement cost (X), we can predict the sales (Y).

2. **Regression Line of X on Y**: This line helps to predict the value of X when Y is known. This is called regression of X on Y.

Example: If we know the sales amount (Y), we can predict how much was likely spent on advertising (X).

Important Note :

- There are always two regression lines when there are two variables.
- The algebraic expression of these regression lines is called as **Regression Equations**.
- Each line is written as a regression equation.
- There will be two regression equations for the two regression lines.
- These equations give the most probable (average) value of one variable based on the other.

Why are There Two Lines?

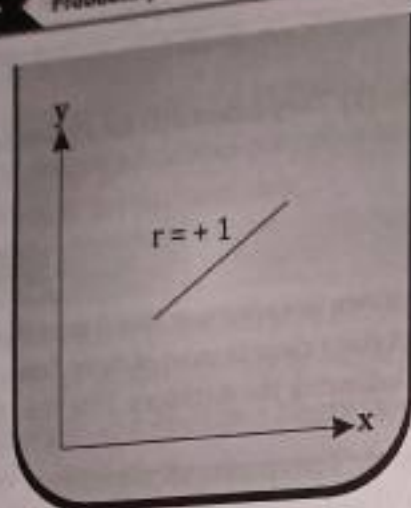
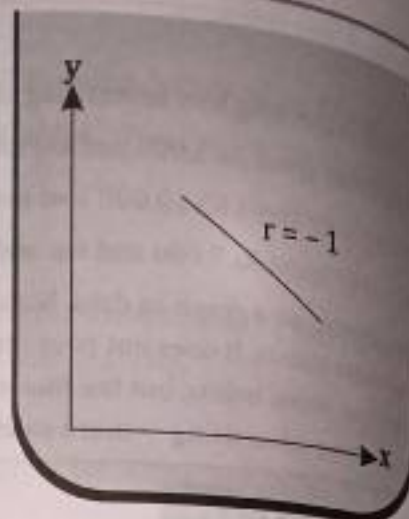
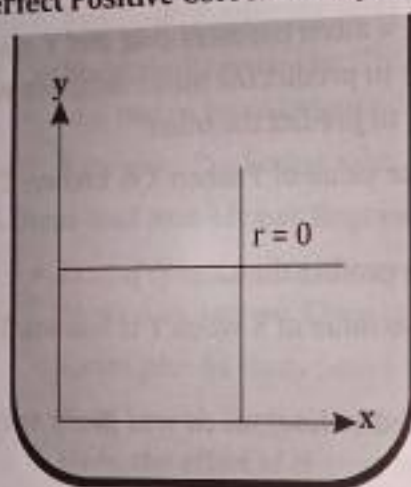
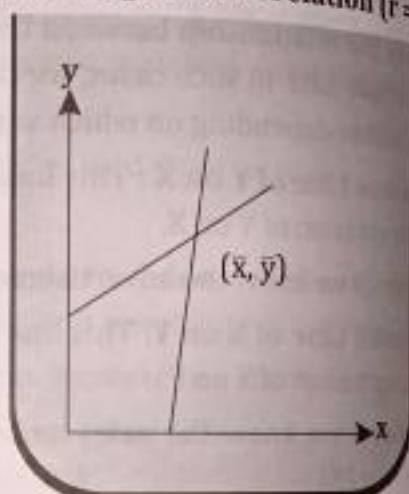
Because either variable can be used to predict the other.

- One line is for predicting Y from X
- The other is for predicting X from Y

These lines are not always the same unless the relationship is perfectly strong.

Understanding the Relationship Between Two Variables Using Regression Lines

The below diagrams show how the two regression lines behave based on the correlation between two variables — such as income and spending, height and weight, or advertisement and sales.

Perfect Positive Correlation ($r = +1$)Perfect Negative Correlation ($r = -1$)No Correlation ($r = 0$)When r is between -1 and $+1$ (Any Normal Correlation)1. Perfect Positive Correlation ($r = +1$)

- When two regression lines are drawn on a graph, and the correlation between X and Y is perfectly positive ($r = +1$), Both lines lie exactly on top of each other.
- This means the two variables increase together in a fixed way, and the two regression lines become one single straight line rising upwards.

2. Perfect Negative Correlation ($r = -1$)

- When two regression lines are drawn on a graph, and the correlation is perfectly negative ($r = -1$), both lines still lie on top of each other, but the line slopes downward.
- This means when one variable increases, the other decreases in a fixed way. Again, the two regression lines become one single straight line, but going downwards.

3. No Correlation ($r = 0$)

- When two regression lines are drawn on a graph, and the correlation is zero ($r = 0$), the two lines meet at right angles (90°).
- This means the two variables are not related to each other. There is no clear pattern between X and Y , so the lines cross like a plus sign (+).

4. When r is between -1 and $+1$

- When two regression lines are drawn on a graph, and the correlation is between -1 and $+1$, r is between -1 and $+1$. That point is the intersection of the two lines and the average of the two lines.
- So, no matter what the correlation is, the two regression lines will always intersect at the same point.

- A line
- A line

- The relation between the two variables is:

- If the
- If the

So, the smaller the angle, the weaker the correlation.

5.1.7

Regression

When two variables have a linear relationship, we can express their relationship as a regression equation. This equation is used to predict the value of one variable from the value of the other.

A regression line is the straight line that best fits the data. It is the algebraic form of a regression equation.



Definition: Regression

The Regression Equation is a mathematical equation that expresses the relationship between the dependent variable and the independent variable.

Example: If we know a regression equation, we can use it to predict the value of one variable from the value of the other.

1. Regression Equation

This is used to predict the value of one variable from the value of the other.

Equation: $x = a + by$

- x = the dependent variable
- y = the independent variable
- a = intercept
- b = slope of the line

Example: To estimate the value of x from the value of y , we use the regression equation.

When r is between -1 and $+1$ (Any Normal Case)

- When two regression lines are drawn on a graph, and the correlation is not perfect (that is, r is between -1 and $+1$), the two lines will appear separate but will cross each other at one fixed point.
 - That point is called the mean point $(\bar{x}, \bar{y})$. This point represents the average of X values ($\bar{x}$) and the average of Y values ($\bar{y}$).
 - So, no matter how weak or strong the relationship is (as long as $r \neq \pm 1$), the two regression lines will always intersect at the mean point. From that point:
 - A line drawn downward to the X -axis will touch the mean of X ($\bar{x}$).
 - A line drawn across to the Y -axis will touch the mean of Y ($\bar{y}$).
 - The relationship can be understood by looking at the angle between the two regression lines:
 - If the two lines are close together, the correlation is strong (r is close to $+1$ or -1).
 - If the two lines are wide apart, the correlation is weak (r is close to 0).
- So, the smaller the angle between the lines, the stronger the relationship. The wider the angle, the weaker the relationship.

5.17 Regression Equation

When two variables have a linear relationship (that is, they change together in a straight-line pattern), we can express their relationship using a mathematical formula. This formula is called a **regression equation**. This equation helps to predict the value of one variable if the value of the other is known. A regression line is the straight line that best fits the plotted data points. A regression equation is the algebraic form of a regression line. Since there are two variables (X and Y), there are two regression equations:

Definition: Regression Equation

The Regression Equation is the algebraic expression of the regression lines. It is used to predict the values of the dependent variable from the given values of independent variables.

Example: If we know a person's advertising budget (X), we can predict the likely sales amount (Y) using a regression equation.

1. Regression Equation of X on Y

This is used to predict X when Y is known.

Equation: $x = a + by$

- x = the dependent variable (value we want to predict)
- y = the independent variable (known value)
- a = intercept (constant)
- b = slope of the regression line

Example: To estimate advertisement cost (X) based on known sales (Y)

2. Regression Equation of Y on X

This is used to predict Y when X is known.

Equation: $y = a + bx$

- y = the dependent variable
- x = the independent variable
- a = intercept (constant)
- b = slope of the regression line

Example: To estimate expected sales (Y) based on the given advertisement cost (X)

5.1.8 Differences between Correlation and Regression

Correlation	Regression
1. It measures the degree and direction of relationship between the variables.	1. It measures the nature and extent of relationship between two or more variables.
2. Shows how strongly two variables are related, and in which direction (positive or negative).	2. Shows how one variable changes when the other changes (used for prediction).
3. Correlation coefficient is a relative measure.	3. Regression is an absolute measure.
4. It need not indicate the cause and effect relationship between the variables.	4. It indicates the cause and effect relationship between the variables.
5. Correlation coefficient is symmetric i.e., $r_{xy} = r_{yx}$	5. Regression coefficient are not symmetric i.e., $b_{xy} \neq b_{yx}$.
6. Expression of the relationship ranges between -1 to +1.	6. Expressed as an equation like $y = a + bx$.
7. Can sometimes show a meaningless (nonsense) correlation.	7. Regression always has a logical connection and purpose.
8. Used only to measure direction and strength of a relationship.	8. Used to predict and estimate values based on known data.

5.2 Linear Regression

In real-life situations, many events and quantities are related to each other. For example:

- The price of a product and the demand for it.
- Hours of study and marks obtained.
- Advertising expenditure and sales revenue.

Studying such relationships helps in understanding patterns and predicting future outcomes.

In statistics, the process of establishing a relationship between two or more variables is called **regression analysis**. The simplest form of regression is linear regression, which studies the relationship between two variables in the form of a straight line.

Meaning and Definition: Linear Regression

Meaning and Definition: Linear Regression

Linear regression is a statistical method used to measure and analyse the relationship between two variables:

- **Dependent Variable (y)** - the outcome or effect we are trying to predict.
- **Independent Variable (x)** - the factor that influences or predicts y.

We aim to find the best-fitting straight-line equation that describes how y changes as x changes. This line is known as the regression line. The most common method for finding this line is the Least Squares Method, which minimises the total squared differences between actual values and predicted values.

General Equation of the Regression Line:

where:

The general equation is: $y = a + bx$

- y = dependent variable
- a = intercept (constant)

- x = independent variable
- b = slope of the regression line

12.1 Purpose of Linear Regression

1. The primary purpose of linear regression is to estimate the value of one variable based on the known value of another variable.
2. It is used to understand the nature and strength of the relationship between two variables.
3. It helps in predicting future trends by analysing historical data and fitting a suitable regression line.
4. Linear regression assists in identifying whether the relationship between variables is positive or negative.
5. It is used to determine the rate of change in the dependent variable for each unit change in the independent variable.
6. The method is applied in decision-making processes by providing forecasts that support planning and strategy.
7. It is useful in evaluating the effectiveness of one factor on another, such as the impact of advertising expenses on sales.
8. Linear regression helps in simplifying complex data relationships into an easy-to-understand mathematical equation.

Example: Example Scenarios Where Linear Regression is Used

1. **Sales Forecasting**: Predicting sales for upcoming months based on past sales data.
2. **Population Studies**: Estimating future population from historical growth records.
3. **Economics**: Analysing how GDP changes with investment or export levels.
4. **Education**: Predicting students' exam scores from their study hours.
5. **Agriculture**: Estimating crop yield based on rainfall or fertilizer used.
6. **Business Analysis**: Finding the relationship between advertisement expenses and revenue.

Example Simple Case

Study Hours (x)	Marks Scored (y)
2	40
3	50
4	60
5	65
6	70

Using the least squares method, we find: $a = 25$ (intercept)
 $b = 8$ (slope)

So the regression equation is: $y = 25 + 8x$

If a student studies 7 hours, predicted marks: $y = 25 + 8(7) = 81$

5.2.3 Characteristics of Linear Regression

1. Linear regression assumes that there is a straight-line relationship between the independent variable (X) and the dependent variable (Y).
2. It involves two variables, where one is the dependent variable (y) and the other is the independent variable (x).
3. The regression line is determined using the least squares method, which minimises the squared differences between the actual values and the predicted values.
4. For a given dataset, there is only one regression line that best fits the data.
5. The slope of the regression line shows the direction of the relationship, which can be positive or negative.
6. The regression line always passes through the point $(\bar{x}, \bar{y})$, which represents the mean of x and the mean of y.
7. The regression equation can be used to predict the values of y for given values of x within the observed range.
8. The regression line is affected by extreme values or outliers, which can distort the results.

5.2.4 Advantages and Limitations of Linear Regression**Advantages of Linear Regression**

1. Linear regression is simple to understand and easy to apply for analysing the relationship between two variables.
2. It provides a clear mathematical equation that can be used to predict the value of the dependent variable for a given value of the independent variable.
3. The method uses all the data points in the dataset to determine the best-fitting line, which is statistically reliable when assumptions are met.
4. The results of linear regression, such as the slope and intercept, can be interpreted in a straightforward manner to understand the nature and strength of the relationship.
5. It is widely used in various fields such as economics, business, engineering, and science for forecasting and trend analysis.

Limitations of Linear Regression

1. Linear regression assumes a linear relationship between the variables, which may not always be the case.
2. The method is sensitive to outliers, which can significantly affect the regression line.
3. It requires the data to be normally distributed, which may not be the case for all datasets.
4. Linear regression is a simple model and may not capture the underlying complexity of the relationship.
5. The method only provides a single best-fitting line, which may not be the most appropriate for all datasets.

5.2.5 Method of Least Squares

The Least Squares Method is used to find the best-fitting line to a set of data points. It minimises the sum of the squared differences between the actual values and the predicted values.

Where:

- y = Predicted value
- x = Value of independent variable
- a is the intercept
- b is the slope (change in y for a unit change in x)

This method can be used to find the best-fitting line for a set of data points. The variables 'x' and 'y' can be sales and profit, respectively. To calculate a and b, the following formulas are used:

$$1. na + b\sum x = \sum y$$

This is the first equation.

$$\bullet n = \text{Number of data points}$$

$$\bullet \sum x = \text{Sum of x values}$$

$$\bullet \sum y = \text{Sum of y values}$$

$$2. a\sum x + b\sum x^2 = \sum xy$$

This is the second equation.

$$\bullet \sum x^2 = \text{Sum of squared x values}$$

$$\bullet \sum xy = \text{Sum of the product of x and y values}$$

To make calculation easier, the total $(\sum x)$ becomes 100.

Limitations of Linear Regression

1. Linear regression assumes a straight-line relationship between variables, which may not hold true if the relationship is actually non-linear.
2. The method is sensitive to extreme values or outliers, which can significantly affect the accuracy of the regression line.
3. It requires that the independent variable and dependent variable have a consistent relationship, which may not be the case in real-world data with fluctuations.
4. Linear regression is limited to making predictions within the range of observed data and may give unreliable results for values outside this range.
5. The method cannot fully explain the relationship between variables if other influencing factors are not included in the model.

1.4.5 Method of Least Squares

The Least Squares Method is a statistical technique used to fit the best possible straight-line equation to a set of data points. It works by minimising the sum of the squares of the vertical differences between the actual values of y and the predicted values from the line.

The equation to fit a straight line trend is $y = a + bx$

Where:

- y = Predicted value of the dependent variable. (like sales, marks, population, etc.)
- x = Value of the independent variable (like year, study hours, time, price), etc
- a is the intercept (value of y when $x = 0$)
- b is the slope or rate of change (how much y increases or decreases for every one unit increase in x)

This method can be used for any type of data — ' x ' can be time, price, temperature, study hours, etc., and ' y ' can be sales, marks, production, or any quantity to be predicted.

To calculate a and b , we need two equations. These are called the **normal equations**.

$$1. na + b\sum x = \sum y$$

This is the first equation.

- n = Number of observations.
- $\sum x$ = Sum of all x values.
- $\sum y$ = Sum of all y values.

$$2. a\sum x + b\sum x^2 = \sum xy$$

This is the second equation.

- $\sum x^2$ = Sum of the squares of all x (x squared)
- $\sum xy$ = Sum of the products of x and y .

To make calculations simple and reduce the number of steps, we adjust the ' x ' values so that their total ($\sum x$) becomes 0.

• When $\Sigma x = 0$:

- The normal equations becomes much easier to solve.
- The formula for calculating 'a' becomes : $a = \Sigma y / n$
- The formula for calculating 'b' becomes : $b = \Sigma xy / \Sigma x^2$
- This saves time and makes the math simpler.

Example How to calculate $\Sigma x = 0$ if odd number of observations are given.

Years (X)	$x = \text{Year} - \bar{x} = X - \bar{x}$ (A.M of years)
2001	$(2001 - 2004) = -3$
2002	$(2002 - 2004) = -2$
2003	$(2003 - 2004) = -1$
2004	$(2004 - 2004) = 0$
2005	$(2005 - 2004) = 1$
2006	$(2006 - 2004) = 2$
2007	$(2007 - 2004) = 3$
$\Sigma X = 14,028$	$\Sigma x = 0$

In this example, the number of observation is odd ($n=7$)

$$\begin{aligned}\bar{x} &= \frac{\Sigma X}{n} \\ &= \frac{2001 + 2002 + 2003 + 2004 + 2005 + 2006 + 2007}{7} \\ &= \frac{14028}{7} = 2004\end{aligned}$$

Example How to calculate $\Sigma x = 0$ if even number of observations are given.

Years (X)	2001	2002	2003	2004	2005	2006	2007	2008
-----------	------	------	------	------	------	------	------	------

Years (X)	x
2001	-7
2002	-5
2003	-3
2004	-1
2005	1
2006	3
2007	5
2008	7
$\Sigma X = 16036$	$\Sigma x = 0$

In this example, the number of observation is even ($n=8$)

Calculation of $\bar{x}$

$$\begin{aligned}\bar{x} &= \frac{\Sigma X}{n} \\ &= \frac{2001 + 2002 + 2003 + 2004 + 2005 + 2006 + 2007 + 2008}{8} = 2004.5\end{aligned}$$

$$x = \frac{\text{year} - \text{Arithmetic mean of years}}{\frac{1}{2} \text{ of interval}}$$

$$2001 = \frac{2001 - 2004.5}{0.5} = \frac{-3.5}{0.5} = -7$$

$$2002 = \frac{2002 - 2004.5}{0.5} = \frac{-2.5}{0.5} = -5$$

Problems on Least Square Method or Linear Method or First Degree Equation Method

Illustration 1

From the following table fit a straight line trend by least square method and estimate the production for the year 2012.

Years(X)	2005	2006	2007	2008	2009	2010	2011
Production(y)	12	10	14	11	13	15	16

Solution

Years (X)	Prod. (y)	$x = X - \bar{X}$	xy	x^2	Trend values ($y_c = a + bx$)
2005	12	(2005 - 2008) -3	-36	9	$13 + (0.75 \times -3) = 10.75$
2006	10	(2006 - 2008) -2	-20	4	$13 + (0.75 \times -2) = 11.5$
2007	14	(2007 - 2008) -1	-14	1	$13 + (0.75 \times -1) = 12.25$
2008	11	(2008 - 2008) 0	0	0	$13 + (0.75 \times 0) = 13$
2009	13	(2009 - 2008) 1	13	1	$13 + (0.75 \times 1) = 13.75$
2010	15	(2010 - 2008) 2	30	4	$13 + (0.75 \times 2) = 14.5$
2011	16	(2011 - 2008) 3	48	9	$13 + (0.75 \times 3) = 15.25$
$\Sigma X = 14056$	$\Sigma y = 91$	$\Sigma x = 0$	$\Sigma xy = 21$	$\Sigma x^2 = 28$	$\Sigma y_c = 91$

Calculation of $\bar{x}$ (Odd years are given) : $\bar{x} = \frac{\Sigma X}{n} = \frac{14056}{7} = 2008$

a. $a = \frac{\Sigma y}{n} = \frac{91}{7} = 13$

b. $b = \frac{\Sigma xy}{\Sigma x^2} = \frac{21}{28} = 0.75$

Using the Least Squares Method, the fitted trend equation is:

$$y = a + bx$$

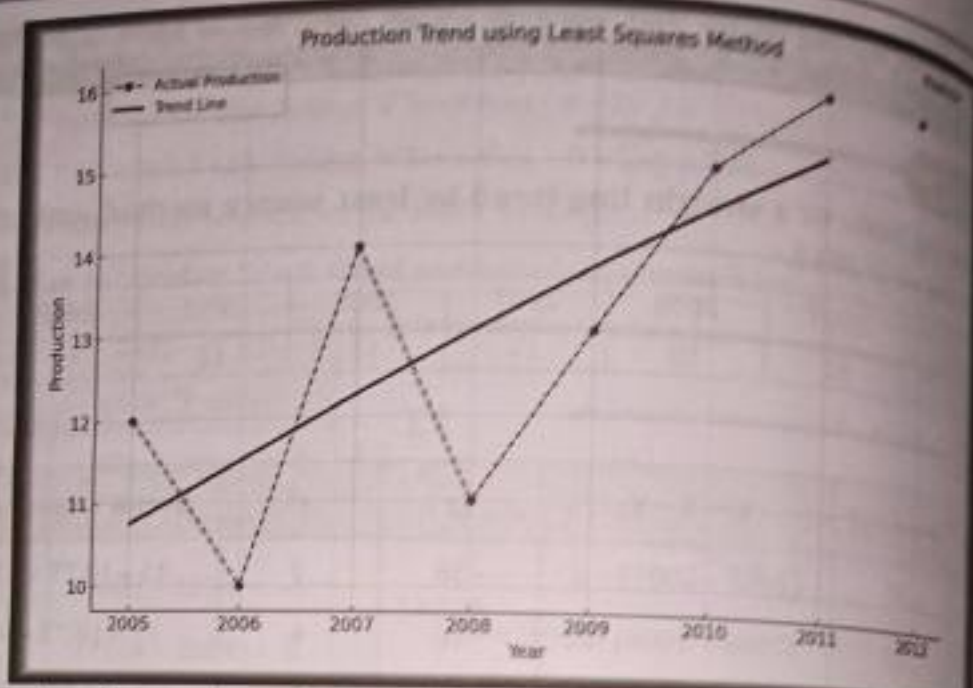
$$y = 13 + 0.75x$$

Here, x is the number of years from the base year 2008 (i.e., $x = \text{Year} - 2008$).

c. Estimate the value for the year 2012 ($x = 2012 - 2008 = 4$)

$$y_c = a + bx = 13 + (0.75 \times 4) = 13 + 3 = 16.$$

The trend values for each year show a gradual increase in production. Based on this trend, the estimated production for the year 2012 ($x = 4$) is 16 units.



The graph displays the actual production values as dotted black lines and the fitted trend line using the least squares method as a solid dark line. Each point on the dotted line represents the production for that year, while the solid line shows the general trend over time. The prediction for the year 2012 is also marked clearly, helping to understand how the trend continues beyond the available data. This visual makes it easy to compare actual data with the estimated trend and see the rate of production growth.

Illustration 2

Fit a straight line by the method of least squares from the data given below, and find the trend values and predict the sales for the year 2008.

Year (X)	1998	1999	2000	2001	2002	2003	2004
Sales (y)	15	18	20	30	39	40	44

Solution

Year (X)	Sales (y)	$x = X - \bar{X}$	x^2	xy	Trend Value
1998	15	-7	49	-105	11.52
1999	18	-5	25	-90	13.00
2000	20	-3	9	-60	14.00
2001	30	-1	1	-30	15.00
2002	39	1	1	39	16.00
2003	40	3	9	120	17.00
2004	44	5	25	220	18.00
2005	50	7	49	350	19.00
$\Sigma X = 16012$	$\Sigma y = 256$	$\Sigma x = 0$	$\Sigma x^2 = 168$	$\Sigma xy = 444$	$\Sigma y = 256$

Calculation of $\bar{x}$ (Even

$$\bar{x} = \frac{1}{n} \sum x$$

$$1998 = \frac{1}{n} \sum x$$

$$2000 = \frac{2}{n} \sum x$$

$$2002 = \frac{2}{n} \sum x$$

$$2004 = \frac{2}{n} \sum x$$

The equation to fit a s

$$a = \frac{\Sigma y}{n}$$

The fitted line

$$y = a + bx$$

The trend values are

$$x = -7 \quad y = a$$

$$y = 3$$

$$y = 3$$

$$y = 3$$

$$= 3$$

$$x = -5 \quad y = 3$$

$$= 3$$

$$x = -3 \quad y = 3$$

$$= 3$$

$$x = -1 \quad y = 3$$

$$= 3$$

$$= 3$$

The estimated sales

$$2008 = \frac{2}{n} \sum x$$

$$x = 13, \quad y = 3$$

$$= 3$$

$$= 3$$

Estimated sal

Calculation of x (Even years are given)

$$\bar{x} = \frac{\sum X}{n} = \frac{16012}{8} = 2001.5$$

$$x = \frac{\text{Year} - \text{AM of years}}{\frac{1}{2} \text{ of interval}}$$

$$1998 = \frac{1998 - 2001.5}{0.5} = \frac{-3.5}{0.5} = -7$$

$$1999 = \frac{1999 - 2001.5}{0.5} = \frac{-2.5}{0.5} = -5$$

$$2000 = \frac{2000 - 2001.5}{0.5} = \frac{-1.5}{0.5} = -3$$

$$2001 = \frac{2001 - 2001.5}{0.5} = \frac{-0.5}{0.5} = -1$$

$$2002 = \frac{2002 - 2001.5}{0.5} = \frac{0.5}{0.5} = 1$$

$$2003 = \frac{2003 - 2001.5}{0.5} = \frac{1.5}{0.5} = 3$$

$$2004 = \frac{2004 - 2001.5}{0.5} = \frac{2.5}{0.5} = 5$$

$$2005 = \frac{2005 - 2001.5}{0.5} = \frac{3.5}{0.5} = 7$$

The equation to fit a straight line trend is $y = a + bx$

$$a = \frac{\sum y}{n} = \frac{256}{8} = 32 \quad b = \frac{\sum xy}{\sum x^2} = \frac{444}{168} = 2.64$$

The fitted linear trend equation is $y = a + bx$

$$y = 32 + 2.64x$$

The trend values are obtained by estimating the values of x

$$x = -7 \quad y = a + bx$$

$$y = 32 + 2.64x$$

$$y = 32 + 2.64(-7) \\ = 32 - 18.48 = 13.52$$

$$x = -5 \quad y = 32 + 2.64(-5)$$

$$= 32 - 13.2 = 18.8$$

$$x = -3 \quad y = 32 + 2.64(-3)$$

$$= 32 - 7.92 = 24.08$$

$$x = -1 \quad y = 32 + 2.64(-1)$$

$$= 32 - 2.64 = 29.36$$

$$x = 1 \quad y = 32 + 2.64(1)$$

$$= 32 + 2.64 = 34.64$$

$$x = 5 \quad y = 32 + 2.64(5)$$

$$= 32 + 13.2 = 45.2$$

$$x = 3 \quad y = 32 + 2.64(3)$$

$$= 32 + 7.92 = 39.92$$

$$x = 7 \quad y = 32 + 2.64(7)$$

$$= 32 + 18.48 = 50.48$$

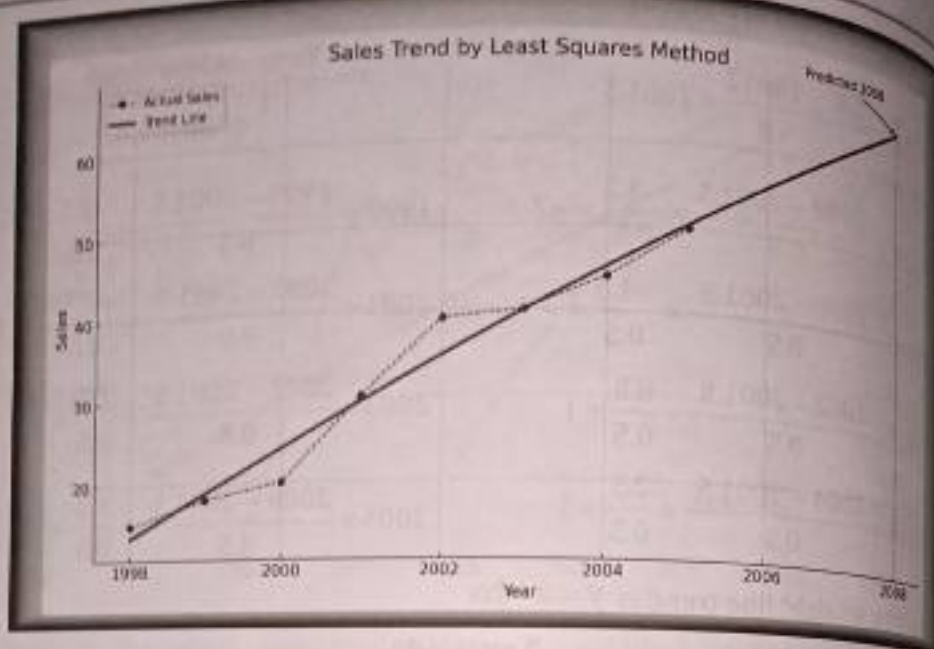
The estimated sales for the year 2008

$$2008 = \frac{2008 - 2001.5}{0.5} = \frac{6.5}{0.5} = 13$$

$$y = 32 + 2.64(13)$$

$$= 32 + 34.32 = 66.32$$

Estimated sales for the year 2008 is ₹66.32.



The graph shows the trend of actual sales from 1998 to 2005 using dotted black lines and estimated trend line using a bold dark solid line. The dotted points represent real sales figures each year. The solid trend line, calculated by the least squares method, shows the general direction of sales over time. The point for the year 2008 is marked on the extended trend line, showing estimated sales value of ₹66.32.

Illustration 3

Determine the trend values by the method of least squares from the following data.

Year (X)	1999	2000	2001	2002	2003	2004	2005
Sales (y)	75	67	68	65	50	54	41

Solution

Year (X)	Sales (y)	$x = X - \bar{X}$	x^2	xy	Trend value
1999	75	-3	9	-225	75.94
2000	67	-2	4	-134	72.43
2001	68	-1	1	-68	65.21
2002	65	0	0	0	60
2003	50	1	1	50	54.79
2004	54	2	4	108	49.57
2005	41	3	9	123	44.36
$\Sigma X = 14014$	$\Sigma y = 420$	$\Sigma x = 0$	$\Sigma x^2 = 28$	$\Sigma xy = -146$	

Calculation of $\bar{x}$ (Odd years)

$$\bar{x} = \frac{\Sigma x}{n}$$

The equation to obtain a is

$$a = \frac{\Sigma y}{n}$$

$$y_c = 60 +$$

The trend values are obtained as

$$y = a + b$$

$$x = -3 \quad = 60 +$$

$$= 60 +$$

$$= 60 +$$

$$x = -2 \quad y = 60 +$$

$$= 60 +$$

$$y = 60 +$$

$$x = -1 \quad = 60 +$$

$$= 60 +$$



The graph shows the actual sales (plotted as a thick dark line), but the trend line is from 1999 to 2005.

Calculation of $\bar{x}$ (Odd years are given)

$$\bar{x} = \frac{\sum X}{n} = \frac{14014}{7} = 2002$$

The equation to obtain a straight line trend is $y = a + bx$

$$a = \frac{\sum y}{n} = \frac{420}{7} = 60$$

$$b = \frac{\sum xy}{\sum x^2} = \frac{-146}{28} = -5.2143$$

$$y_t = 60 + (-5.2143)x \Rightarrow 60 - 5.2143x$$

The trend values are obtained by estimating the values of x

$$\begin{aligned} x = -3 \quad y &= a + bx \\ &= 60 + (-5.2143)(-3) \\ &= 60 + 15.6429 = 75.63 \end{aligned}$$

$$\begin{aligned} x = -2 \quad y &= 60 + (-5.2143)(-2) \\ &= 60 + 10.4286 = 70.43 \end{aligned}$$

$$\begin{aligned} x = -1 \quad y &= 60 + (-5.2143)(-1) \\ &= 60 + 5.2143 = 65.21 \end{aligned}$$

$$x = 0 \quad y = 60 + (-5.2143)(0) = 60$$

$$\begin{aligned} x = 1 \quad y &= 60 + (-5.2143)(1) \\ &= 60 - 5.2143 = 54.79 \end{aligned}$$

$$\begin{aligned} x = 2 \quad y &= 60 + (-5.2143)(2) \\ &= 60 - 10.4286 = 49.57 \end{aligned}$$

$$\begin{aligned} x = 3 \quad y &= 60 + (-5.2143)(3) \\ &= 60 - 15.6429 = 44.36 \end{aligned}$$



The graph shows the actual sales values (plotted as a dotted black line) and the estimated trend line (plotted as a thick dark solid line) using the least squares method. The actual sales fluctuate over the years, but the trend line shows a clear downward direction. This indicates a steady decrease in sales from 1999 to 2005.

Illustration 4

Fit a straight line trend under square method to the following data. Estimate the production for the year 2007.

Years (X)	2000	2001	2002	2003	2004	2005
Production (In 000 tons) (y)	35	42	44	48	46	49

Solution

Years (X)	Production (In 000 tons) (y)	$x = X - \bar{X}$	x^2	xy	Trend values
2000	35	-3	9	-105	38.13
2001	42	-2	4	-84	40.42
2002	44	-1	1	-44	42.71
2003	48	0	0	0	45
2004	46	1	1	46	47.29
2005	49	2	4	98	49.58
2006	51	3	9	153	51.87
$\Sigma X = 14021$	$\Sigma y = 315$	$\Sigma x = 0$	$\Sigma x^2 = 28$	$\Sigma xy = 64$	$\Sigma y_c = 315$

Calculation of $\bar{x}$ (Odd years are given)

$$\bar{x} = \frac{\Sigma X}{n} = \frac{14021}{7} = 2003$$

The equation to fit a straight line trend is $y = a + bx$

$$a = \frac{\Sigma y}{n} = \frac{315}{7} = 45 \quad b = \frac{\Sigma xy}{\Sigma x^2} = \frac{64}{28} = 2.29$$

$$Y_c = 45 + 2.29x$$

The trend values are obtained by estimating the values of x

$x = -3$	$y = a + bx$ $= 45 + 2.29(-3)$ $= 45 - 6.87 = 38.13$	$x = 0$	$y = 45 + 2.29(0) = 45$
$x = -2$	$y = 45 + 2.29(-2)$ $= 45 - 4.58 = 40.42$	$x = 1$	$y = 45 + 2.29(1) = 47.29$
$x = -1$	$y = 45 + 2.29(-1)$ $= 45 - 2.29 = 42.71$	$x = 2$	$y = 45 + 2.29(2) = 49.58$
		$x = 3$	$y = 45 + 2.29(3) = 51.87$

The estimated production for the year 2007

Value of x for 2007 = $2007 - 2003 = 4$

$$y = 45 + 2.29(4) = 45 + 9.16$$

$$y = 54.16$$

Estimate

The graph shows a solid dark trend line. A point marked on the line indicates the overall growth.

Illustration

The production

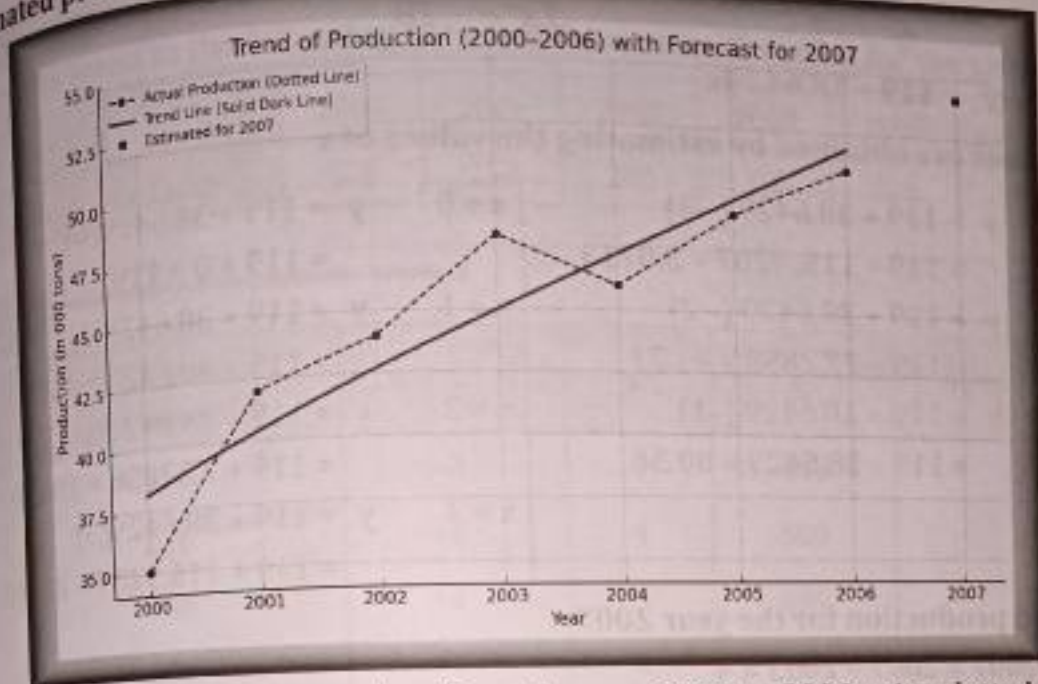
Year (X)
Production (y)
Estimate the

Solution

Year (X)
2000
2001
2002
2003
2004
2005
2006
$\Sigma X = 14021$
Calculation of

The equation to

Estimated production for the year 2007 is 54.16 tons.



The graph shows actual production values from the year 2000 to 2006 using dotted black lines. A solid dark trend line represents the average pattern calculated using the least squares method. The point marked for 2007 shows the predicted production, which lies on the trend line, continuing the overall growth pattern.

Illustration 5

The production of a company in thousands of rupees for the year 2000 through 2006 are given:

Year (X)	2000	2001	2002	2003	2004	2005	2006
Production (y)	32	47	65	92	132	190	275

Estimate the production for the year 2007 using the equation of straight line.

Solution

Year (X)	Production (y)	$x = X - \bar{X}$	x^2	xy	Trend values (y_t)
2000	32	-3	9	-96	3.07
2001	47	-2	4	-94	41.71
2002	65	-1	1	-65	80.36
2003	92	0	0	0	119
2004	132	1	1	132	157.64
2005	190	2	4	380	196.29
2006	275	3	9	825	234.93
$\Sigma x = 14021$	$\Sigma y = 833$	$\Sigma x = 0$	$\Sigma x^2 = 28$	$\Sigma xy = 1082$	$\Sigma y_t = 833$

Calculation of $\bar{x}$ (Odd years are given)

$$\bar{x} = \frac{\Sigma X}{n} = \frac{2000 + 2001 + 2002 + 2003 + 2004 + 2005 + 2006}{7} = \frac{14021}{7} = 2003$$

The equation to fit a straight line trend is $y = a + bx$

If
$$a = \frac{\Sigma y}{n} = \frac{833}{7} = 119; \quad b = \frac{\Sigma xy}{\Sigma x^2} = \frac{1082}{28} = 38.6429$$

$$y_c = 119 + 38.6429x$$

The trend values are obtained by estimating the values of x

$$x = -3 \quad y = 119 + 38.6429(-3) \\ = 119 - 115.9287 = 3.0713$$

$$x = -2 \quad y = 119 + 38.6429(-2) \\ = 119 - 77.2858 = 41.71$$

$$x = -1 \quad y = 119 + 38.6429(-1) \\ = 119 - 38.6429 = 80.36$$

$$x = 0 \quad y = 119 + 38.6429(0) \\ = 119 + 0 = 119$$

$$x = 1 \quad y = 119 + 38.6429(1) \\ = 119 + 38.6429 = 157.64$$

$$x = 2 \quad y = 119 + 38.6429(2) \\ = 119 + 77.2858 = 196.29$$

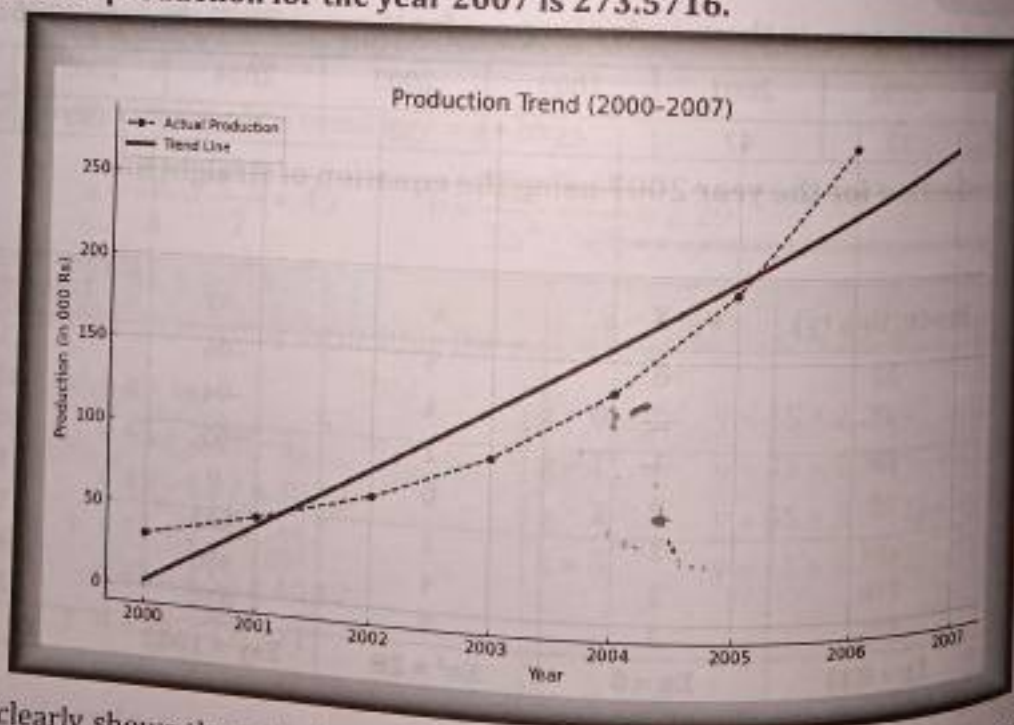
$$x = 3 \quad y = 119 + 38.6429(3) \\ = 119 + 115.9287 = 234.93$$

The Estimated production for the year 2007

$$\text{Value of } x \text{ for 2007} = 2007 - 2003 = 4$$

$$x = 4 \quad y = 119 + 38.6429x \\ = 119 + 38.6429(4) \\ = 119 + 154.5716 \\ y = 273.5716$$

∴ Estimated production for the year 2007 is 273.5716.



The graph clearly shows the actual production values as dotted black points for the years 2000-2006. The solid dark line represents the trend line obtained using the least squares method. It smoothly through the central direction of the data. The point for the year 2007 on the trend line shows the estimated production, which is 273.57.

Illustration 6
Fit a straight line

Year
Sales of product

Solution

Year (X)	
2005	
2006	
2007	
2008	
2009	
2010	
2011	
EX = 14056	

Calculation of x

The equation to

The trend values

$$x = -3$$

$$x = -2$$

$$x = -1$$

Illustration 6

Fit a straight line trend to the following data and determine the sales for the year 2014.

Year (X)	2005	2006	2007	2008	2009	2010	2011
Sales of product 'y' (in ₹ lakhs)	200	250	280	350	325	400	430

Solution

Year (X)	Sales of product (in ₹ lakhs) (y)	$x = X - \bar{x}$	x^2	xy	Trend values (y_c)
2005	200	-3	9	-600	208.39
2006	250	-2	4	-500	245.36
2007	280	-1	1	-280	282.32
2008	350	0	0	0	319.29
2009	325	1	1	325	356.25
2010	400	2	4	800	393.21
2011	430	3	9	1290	430.18
$\Sigma = 14056$	$\Sigma y = 2235$	$\Sigma x = 0$	$\Sigma x^2 = 28$	$\Sigma xy = 1035$	$\Sigma y_c = 2235$

Calculation of $\bar{x}$ (Odd years are given)

$$\bar{x} = \frac{\Sigma X}{n} = \frac{2005 + 2006 + 2007 + 2008 + 2009 + 2010 + 2011}{7} = \frac{14056}{7} = 2008$$

The equation to fit a straight line trend is $y = a + bx$

$$a = \frac{\Sigma y}{n} = \frac{2235}{7} = 319.2857; \quad b = \frac{\Sigma xy}{\Sigma x^2} = \frac{1035}{28} = 36.9643$$

$$y_c = 319.2857 + 36.9643x$$

The trend values are obtained by estimating the values of x

$$\begin{aligned} x = -3 \quad y &= 319.2857 + 36.9643(-3) \\ &= 319.2857 - 110.8929 = 208.39 \\ x = -2 \quad y &= 319.2857 + 36.9643(-2) \\ &= 319.2857 - 73.9286 = 245.36 \\ x = -1 \quad y &= 319.2857 + 36.9643(-1) \\ &= 319.2857 - 36.9643 = 282.32 \end{aligned}$$

$$\begin{aligned} x = 0 \quad y &= 319.2857 + 36.9643(0) \\ &= 319.2857 + 0 = 319.2857 \\ x = 1 \quad y &= 319.2857 + 36.9643(1) \\ &= 319.2857 + 36.9643 = 356.25 \\ x = 2 \quad y &= 319.2857 + 36.9643(2) \\ &= 319.2857 + 73.9286 = 393.21 \\ x = 3 \quad y &= 319.2857 + 36.9643(3) \\ &= 319.2857 + 110.8929 = 430.18 \end{aligned}$$

years 2000 to
mod. It passes
the trend line

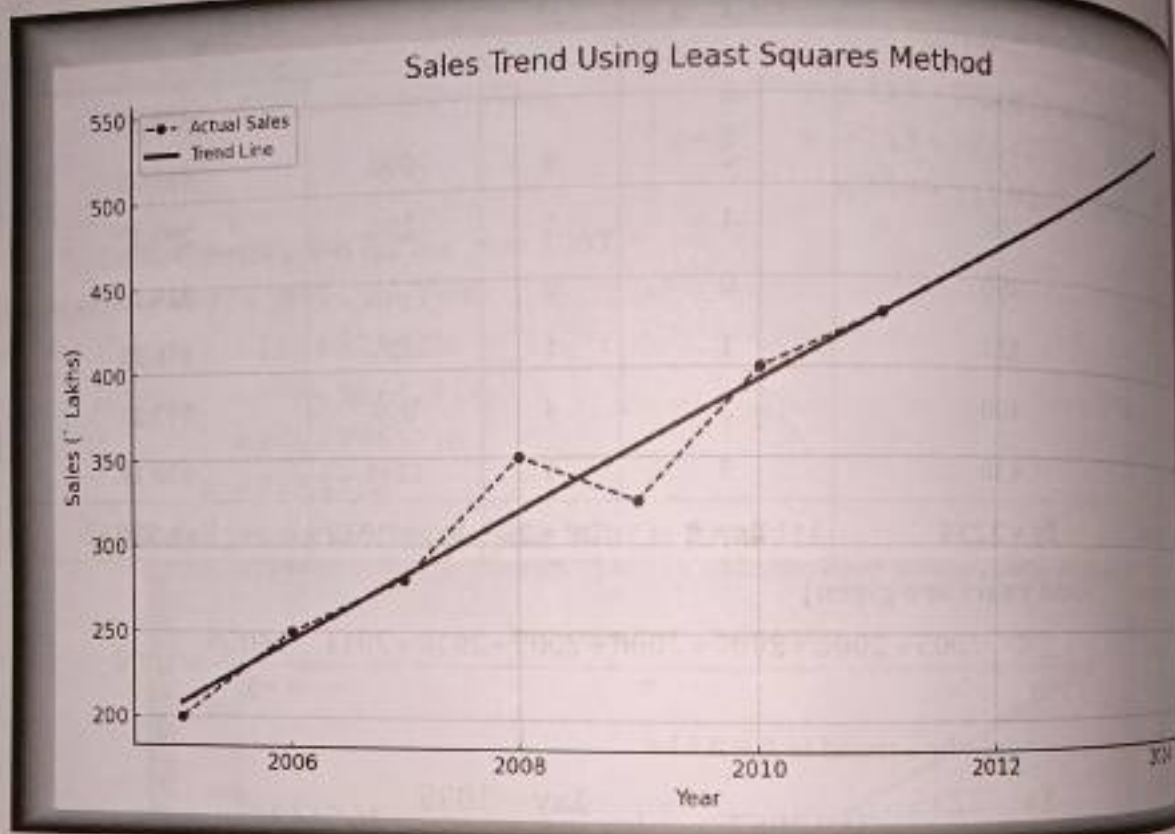
The Estimated sales for the year 2014

Value of x for 2014 = $2014 - 2008 = 6$

$$\begin{aligned} x = 6 \quad y &= 319.2857 + 36.9643x \\ &= 319.2857 + 36.9643(6) \\ &= 319.2857 + 221.7858 \end{aligned}$$

$$y = 541.07$$

Estimated sales for the year 2014 is 541.07 Lakhs.



The graph above displays the sales trend from 2005 to 2011 and includes a projection for 2014 using the least squares method. The dotted line shows the actual sales values each year. The bold solid line represents the fitted trend line, which shows the general direction of sales over time.

Both lines intersect at several points, indicating that the trend equation closely follows the actual data. The upward slope of the trend line confirms that sales have been generally increasing. The predicted sales for 2014, based on this trend, is Rs.541.07 lakhs.

Illustration 7

The number of units of a product exported during 2005 - 2012 are given below. Compute trend values by the method of least squares and prove that $\Sigma(y - y_c) = 0$.

Year (X)	2005	2006	2007	2008	2009	2010	2011
Exports (in lakhs)	24	26	26	32	38	46	42

Solution

Year (X)	Exports (Y)
2005	24
2006	26
2007	26
2008	32
2009	38
2010	46
2011	42
2012	-
$\Sigma X = 16068$	$\Sigma Y = 274$

Calculation of x (Year from 2008)

2005

2006

2007

2008

2009

2010

2011

2012

The equation to fit

y_c

Solution

Year (X)	Exports (₹ in lakhs) (y)	$x = X - \bar{X}$	x^2	xy	Trend values (y_c)
2005	24	-7	49	-168	22.50
2006	26	-5	25	-130	26.07
2007	26	-3	9	-78	29.64
2008	32	-1	1	-32	33.21
2009	38	1	1	38	36.79
2010	46	3	9	138	40.36
2011	42	5	25	210	43.93
2012	46	7	49	322	47.50
$\Sigma = 16068$	$\Sigma y = 280$	$\Sigma x = 0$	$\Sigma x^2 = 168$	$\Sigma xy = 300$	$\Sigma y_c = 280$

Calculation of x (even years are given)

$$\bar{x} = \frac{\Sigma X}{n} = \frac{2005 + 2006 + 2007 + 2008 + 2009 + 2010 + 2011 + 2012}{8}$$

$$= \frac{16068}{8} = 2008.5$$

$$x = \frac{\text{Year} - \text{AM of years}}{\frac{1}{2} \text{ of interval}}$$

$$2005 = \frac{2005 - 2008.5}{0.5} = \frac{-3.5}{0.5} = -7$$

$$2006 = \frac{2006 - 2008.5}{0.5} = \frac{-2.5}{0.5} = -5$$

$$2007 = \frac{2007 - 2008.5}{0.5} = \frac{-1.5}{0.5} = -3$$

$$2008 = \frac{2008 - 2008.5}{0.5} = \frac{-0.5}{0.5} = -1$$

$$2009 = \frac{2009 - 2008.5}{0.5} = \frac{0.5}{0.5} = 1$$

$$2010 = \frac{2010 - 2008.5}{0.5} = \frac{1.5}{0.5} = 3$$

$$2011 = \frac{2011 - 2008.5}{0.5} = \frac{2.5}{0.5} = 5$$

$$2012 = \frac{2012 - 2008.5}{0.5} = \frac{3.5}{0.5} = 7$$

The equation to fit a straight line trend $y = a + bx$

$$a = \frac{\Sigma y}{n} = \frac{280}{8} = 35; \quad b = \frac{\Sigma xy}{\Sigma x^2} = \frac{300}{168} = 1.786$$

$$y_c = 35 + 1.786x$$

ion for 2014 using
The bold solid line
ne.
follows the actual
lly increasing. The

elow. Compute the

2011	2012
42	46

The trend values are obtained by estimating the values of x

$$x = -7 \quad y = 35 + 1.786(-7) \\ = 35 - 12.502 = 22.50$$

$$x = -5 \quad y = 35 + 1.786(-5) \\ = 35 - 8.93 = 26.07$$

$$x = -3 \quad y = 35 + 1.786(-3) \\ = 35 - 5.358 = 29.642$$

$$x = -1 \quad y = 35 + 1.786(-1) \\ = 35 - 1.786 = 33.214$$

$$\Sigma(y - y_c) = 0 \Rightarrow \Sigma(280 - 280) = 0$$

$\therefore$ Hence proved.

Illustration 8

Fit a straight line trend to the following by the method of least squares. Assuming the same rate of changes continues state what would be the estimated earnings for the year 2012.

Prove that $\Sigma y_c - \Sigma y = 0$

Year	2005	2006	2007	2008	2009	2010	2011	2012
Earnings (in lakhs)	38	40	65	72	69	60	87	95

Solution

Year (X)	Earning in lakhs (y)	$x = X - \bar{X}$	x^2	xy	Trend values
2005	38	-7	49	-266	40.86
2006	40	-5	25	-200	41.4
2007	65	-3	9	-195	54.74
2008	72	-1	1	-72	62.88
2009	69	1	1	69	69.42
2010	60	3	9	180	76.14
2011	87	5	25	435	88.1
2012	95	7	49	665	91.44
$\Sigma X = 16068$	$\Sigma y = 526$	$\Sigma x = 0$	$\Sigma x^2 = 168$	$\Sigma xy = 616$	$\Sigma y_c = 526$

Calculation of $\bar{x}$ (even years are given)

$$\bar{x} = \frac{\Sigma x}{n} = \frac{2005 + 2006 + 2007 + 2008 + 2009 + 2010 + 2011 + 2012}{8} \\ = \frac{16068}{8} = 2008.5$$

The equation to fit a

$a =$

$y_c =$

The trend values are

$x = -7 \quad y =$

$x = -5 \quad y =$

$x = -3 \quad y =$

$x = -1 \quad y =$

Estimated value for

2016 =

If $x = 15$, $y =$

The estimate

Illustration 9

Fit a straight line to the graph.

Years	Sales (in 000 tons)
-------	---------------------

$$x = \frac{\text{Year-AM of years}}{\frac{1}{2} \text{ of interval}}$$

$$2005 = \frac{2005 - 2008.5}{0.5} = \frac{-3.5}{0.5} = -7$$

$$2006 = \frac{2006 - 2008.5}{0.5} = \frac{-2.5}{0.5} = -5$$

$$2007 = \frac{2007 - 2008.5}{0.5} = \frac{-1.5}{0.5} = -3$$

$$2008 = \frac{2008 - 2008.5}{0.5} = \frac{-0.5}{0.5} = -1$$

$$2009 = \frac{2009 - 2008.5}{0.5} = \frac{0.5}{0.5} = 1$$

$$2010 = \frac{2010 - 2008.5}{0.5} = \frac{1.5}{0.5} = 3$$

$$2011 = \frac{2011 - 2008.5}{0.5} = \frac{2.5}{0.5} = 5$$

$$2012 = \frac{2012 - 2008.5}{0.5} = \frac{3.5}{0.5} = 7$$

The equation to fit a straight line trend is $y = a + bx$

$$a = \frac{\Sigma y}{n} = \frac{526}{8} = 65.75; \quad b = \frac{\Sigma xy}{\Sigma x^2} = \frac{616}{168} = 3.67$$

$$y_c = 65.75 + 3.67x$$

The trend values are obtained by estimating the values of x

$$x = -7 \quad y = 65.75 + 3.67(-7) \\ = 65.75 - 25.69 = 40.06$$

$$x = -5 \quad y = 65.75 + 3.67(-5) \\ = 65.75 - 18.35 = 47.4$$

$$x = -3 \quad y = 65.75 + 3.67(-3) \\ = 65.75 - 11.01 = 54.74$$

$$x = -1 \quad y = 65.75 + 3.67(-1) \\ = 65.75 - 3.67 = 62.08$$

Estimated value for the year 2016

$$2016 = \frac{2016 - 2008.5}{0.5} = 15$$

$$x = 15, \quad y = a + bx \\ = 65.75 + 3.67(15) \\ = 65.75 + 55.05 = 120.8$$

The estimate value for the year 2016 is ₹ 120.8 Lakhs.

Illustration 9

Fit a straight line trend to the following figures by the method of least squares and plot in the graph.

Years	2003	2004	2005	2006	2007	2008
Sales (in 000 tons)	50	60	65	63	68	70

g that the
year 2016.

2012
95

values(y)

40.06

47.4

54.74

62.08

69.42

76.76

84.1

91.44

$\Sigma y_c = 526$

$$x = 1 \quad y = 65.75 + 3.67(1) \\ = 65.75 + 3.67 = 69.42$$

$$x = 3 \quad y = 65.75 + 3.67(3) \\ = 65.75 + 11.01 = 76.76$$

$$x = 5 \quad y = 65.75 + 3.67(5) \\ = 65.75 + 18.35 = 84.1$$

$$x = 7 \quad y = 65.75 + 3.67(7) \\ = 65.75 + 25.69 = 91.44$$

Prove that:

$$\Sigma y_c - \Sigma y = 0$$

$$526 - 526 = 0$$

$$\Sigma y_c - \Sigma y = 0$$

(Hence proved)

Solution

Year (X)	y (Sales in 000 tons)	$x = X - \bar{X}$	x^2	xy	Trend value
2003	50	-5	25	-250	53.95
2004	60	-3	9	-180	57.44
2005	65	-1	1	-65	60.93
2006	63	1	1	63	64.43
2007	68	3	9	204	67.89
2008	70	5	25	350	71.38
$\Sigma X = 12033$	$\Sigma y = 376$	$\Sigma x = 0$	$\Sigma x^2 = 70$	$\Sigma xy = 122$	$\Sigma y = 276$

$$\bar{x} = \frac{\Sigma X}{n} = \frac{2003 + 2004 + 2005 + 2006 + 2007 + 2008}{6} = \frac{12033}{6} = 2005.5$$

$$x = \frac{\text{Year-AM of years}}{\frac{1}{2} \text{ of interval}}$$

$$2003 = \frac{2003 - 2005.5}{0.5} = \frac{-2.5}{0.5} = -5$$

$$2004 = \frac{2004 - 2005.5}{0.5} = \frac{-1.5}{0.5} = -3$$

$$2005 = \frac{2005 - 2005.5}{0.5} = \frac{-0.5}{0.5} = -1$$

$$2006 = \frac{2006 - 2005.5}{0.5} = \frac{0.5}{0.5} = 1$$

$$2007 = \frac{2007 - 2005.5}{0.5} = \frac{1.5}{0.5} = 3$$

$$2008 = \frac{2008 - 2005.5}{0.5} = \frac{2.5}{0.5} = 5$$

Straight line trend is $y = a + bx$

$$a = \frac{\Sigma y}{n} = \frac{376}{6} = 62.67$$

$$b = \frac{\Sigma xy}{\Sigma x^2} = \frac{122}{70} = 1.743$$

$$y_c = 62.67 + 1.743x$$

The trend values are obtained by estimating the values of x

$$x = -5$$

$$y = 62.67 + 1.743(-5) \\ = 62.67 - 8.715 = 53.955$$

$$x = -3$$

$$y = 62.67 + 1.743(-3) \\ = 62.67 - 5.229 = 57.441$$

$$x = -1$$

$$y = 62.67 + 1.743(-1) \\ = 62.67 - 1.743 = 60.93$$

$$x = 1$$

$$y = 62.67 + 1.743(1) \\ = 62.67 + 1.743 = 64.413$$

$$x = 3$$

$$y = 62.67 + 1.743(3) \\ = 62.67 + 5.229 = 67.899$$

$$x = 5$$

$$y = 62.67 + 1.743(5) \\ = 62.67 + 8.715 = 71.385$$

The graph above shows a dotted black line representing the calculated trend.

Each year's trend value is obtained by increasing the performance of the company.

Illustration

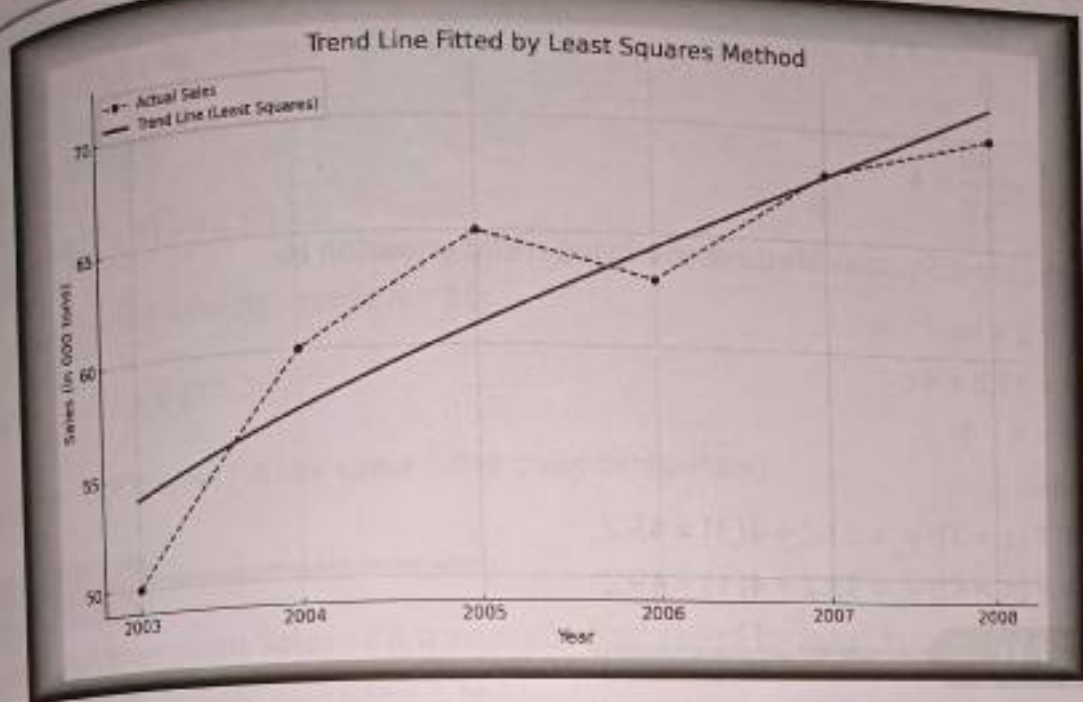
Study Hours
Estimate the

Study Hours
Marks

Solution

Since the number of study hours is increasing, the marks are also increasing. So, $x = X - A$

(X)
2
3
4
5
6
$\Sigma X = 20$



The graph above shows both actual sales and the fitted trend line using the least squares method: The dashed black line represents the actual sales figures from 2003 to 2008. The solid dark line represents the calculated trend values using the equation of a straight line.

Each year's trend value lies close to the actual sales, showing that the data follows a fairly steady increasing pattern. This trend line can help estimate future values and analyze the overall sales performance during the given years.

Illustration 10

Study Hours and Marks : Fit a straight-line trend by least squares for the following data. Estimate the marks when study hours are 7 and 8.

Study Hours (X)	2	3	4	5	6
Marks (y)	25	28	35	38	40

Solution

Since the number of observations ($n = 5$) is odd, take the middle value of $X = 4$ as the origin.

$X = X - A$

(X)	(y)	$x = X - 4$	xy	x^2	Trend values ($y_c = a + bx$)
2	25	-2	-50	4	$33.2 + 4(-2) = 25.2$
3	28	-1	-28	1	$33.2 + 4(-1) = 29.2$
4	35	0	0	0	$33.2 + 4(0) = 33.2$
5	38	1	38	1	$33.2 + 4(1) = 37.2$
6	40	2	80	4	$33.2 + 4(2) = 41.2$
$\Sigma X = 20$	$\Sigma y = 166$	$\Sigma x = 0$	$\Sigma xy = 40$	$\Sigma x^2 = 10$	$\Sigma y_c = 166$

$$a. a = \frac{\sum y}{n} = \frac{166}{5} = 33.2$$

$$b. b = \frac{\sum xy}{\sum x^2} = \frac{40}{10} = 4$$

Using the least Squares Method, the fitted trend equation is:

$$y = a + bx$$

$$y = 33.2 + 4x$$

Here, $x = X - 4$

c. Estimates

$$\text{For } X = 7 \text{ (} x = 3 \text{): } y_c = 33.2 + 4(3) = 45.2$$

$$\text{For } X = 8 \text{ (} x = 4 \text{): } y_c = 33.2 + 4(4) = 49.2$$

Illustration 11

Advertisement and Sales : Fit a straight-line trend by least squares and estimate sales when advertisement expenditure is 50.

Study Hours (X)	2	3	4	5	6
Marks (y)	25	28	35	38	40

Solution

Since the number of observations ($n = 5$) is odd, take the middle value of $X = 4$ as the origin. So, $x = X - 4$. Treat X as Advertisement and y as Sales.

(X)	(y)	$x = X - 4$	xy	x^2	Trend values (y_c)
2	25	-2	-50	4	$33.2 + 4(-2) = 25.2$
3	28	-1	-28	1	$33.2 + 4(-1) = 29.2$
4	35	0	0	0	$33.2 + 4(0) = 33.2$
5	38	1	38	1	$33.2 + 4(1) = 37.2$
6	40	2	80	4	$33.2 + 4(2) = 41.2$
$\Sigma X = 20$	$\Sigma y = 166$	$\Sigma x = 0$	$\Sigma xy = 40$	$\Sigma x^2 = 10$	$\Sigma y_c = 166$

$$a. a = \frac{\sum y}{n} = \frac{166}{5} = 33.2$$

$$b. b = \frac{\sum xy}{\sum x^2} = \frac{40}{10} = 4$$

Using the Least Squares Method, the fitted straight line trend is:
 $y = a + bx = 33.2 + 4x$, where $x = X - 4$.

Estimate the Sales when Advertisement $X = 50$:

$$y = 33.2 + 4(X - 4) = 4X + 17.2$$

Estimate the Sales when Advertisement $X = 50$:

$$y_c = 4(50) + 17.2 = 217.2$$

Estimated Sales : 217.2 units (in the same sales units as the data)

Illustration 12

Temperature and Ice-cream Sales : Fit a straight-line trend by least squares and estimate ice-cream sales when the temperature is 35°C.

Temperature (X in °C)	20	25	30	35	40
Sales (y in Units)	100	130	150	170	200

Solution

Since the number of observations ($n = 5$) is odd, take the middle value of $X = 30$ as the origin.

$$x = X - 30$$

Temperature °C	y(Sales)	$x = X - 30$	xy	x^2	Trend values ($y_c = a + bx$)
20	100	-10	-1000	100	$150 + 5(-10) = 100$
25	130	-5	-650	25	$150 + 5(-5) = 125$
30	150	0	0	0	$150 + 5(0) = 150$
35	170	5	850	25	$150 + 5(5) = 175$
40	200	10	2000	100	$150 + 5(10) = 200$
$\Sigma X = 150$	$\Sigma y = 750$	$\Sigma x = 0$	$\Sigma xy = 1200$	$\Sigma x^2 = 250$	$\Sigma y_c = 750$

$$a = \frac{\Sigma y}{n} = \frac{750}{5} = 150$$

$$b = \frac{\Sigma xy}{\Sigma x^2} = \frac{1200}{250} = 4.8 = 5$$

Using the Least Squares Method, the fitted straight line trend is

$$y = a + bx$$

$$y = 150 + 5x$$

Here,

$$x = X - 30$$

So, in terms of the original X :

$$y = 150 + 5(X - 30) = 5X.$$

Estimate ice cream sales when Temperature $X = 35^\circ\text{C}$:

$$y_c = 5(35) = 175$$

$\therefore$ Estimated Sales = 175 units.

Illustration 13

Years of Experience and Salary : Fit a straight-line trend by least squares and estimate when experience is 9 years.

Experience (X in Years)	2	4	6	8	10
Salary (y in lakhs)	15	20	25	30	35

Solution

Since $n = 6$ (even) and the gap is constant (2 years):

- A.M. of $X = \frac{2+12}{2} = 7$
- Half-interval = 1
- Code: $x = \frac{X-7}{1} = X-7 \Rightarrow x = \{-5, -3, -1, +1, +3, +5\}$ so $\sum x = 0$

X (Years)	y (₹'000)	$x = X - 7$	xy	x^2	Trend values ($y_c = a + bx$)
2	15	-5	-75	25	15.0
4	20	-3	-60	9	20.0
6	25	-1	-25	1	25.0
8	30	1	30	1	30.0
10	35	3	105	9	35.0
12	40	5	200	25	40.0
$\Sigma X = 42$	$\Sigma y = 165$	$\Sigma x = 0$	$\Sigma xy = 175$	$\Sigma x^2 = 70$	$\Sigma y_c = 165$

a. With $\Sigma x = 0$:

$$a = \frac{\Sigma y}{n} = \frac{165}{6} = 27.5;$$

$$b = \frac{\Sigma xy}{\Sigma x^2} = \frac{175}{70} = 2.5$$

b. Fitted line (coded x)

$$y = a + bx = 27.5 + 2.5x, \text{ where } x = X - 7.$$

In original X :

$$y = 10 + 2.5X \text{ (₹'000)}$$

c. Example prediction
 $y_c = 27.5 + 2.5x$

Illustration 14

A company spends different

Advertising X (₹ lakhs)

Sales y (₹ lakhs)

Fit a straight-line trend and estimate sales when advertising is 15 lakhs.

Solution

Step 1: Since intervals are not equal

$$X = \{2, 5, 9, 13, 20\}$$

So we cannot use the method of equal intervals.

We must use normal equations.

Step 2: Prepare Calculations

X
2
5
9
13
20
$\Sigma X = 49$

Step 3: Normal Equations

$$1. \Sigma y = na + b \Sigma x$$

$$360 = 5a + b \Sigma x$$

$$2. \Sigma xy = a \Sigma x + b \Sigma x^2$$

$$4715 = 49a + b \Sigma x^2$$

Step 4: Solve Equations

From (1):

$$a = \frac{360 - b \Sigma x}{5}$$

Substitute in (2):

$$4715 = 49a + b \Sigma x^2$$

Example prediction (say $X = 9$ years): $x = 9 - 7 = 2$
 $\hat{y} = 27.5 + 2.5(2) = 32.5$ (₹'000)

Question 14

A company spends different amounts on advertising (₹) and records corresponding sales (₹):

Advertising X (₹ lakhs)	2	5	9	13	20
Sales Y (₹ lakhs)	25	40	70	95	130

Find the straight-line trend using the least squares method and estimate sales when advertising spend

Solution
 Since intervals are unequal

$X = \{2, 5, 9, 13, \rightarrow 20\}$ gaps are 3, 4, 4, 7.

So we cannot use coding.

We must use normal equations.

Step 1: Prepare Calculation Table

X	Y	X^2	XY
2	25	4	50
5	40	25	200
9	70	81	630
13	95	169	1235
20	130	400	2600
$\Sigma X = 49$	$\Sigma Y = 360$	$\Sigma X^2 = 679$	$\Sigma XY = 4175$

Step 2: Normal Equations

$$1. \quad \Sigma Y = na + b \Sigma X \quad \dots(1)$$

$$360 = 5a + 49b$$

$$2. \quad \Sigma XY = a \Sigma X + b \Sigma X^2 \quad \dots(2)$$

$$4175 = 49a + 679b$$

Step 3: Solve Equations

From (1):

$$a = \frac{360 - 49b}{5}$$

Substitute in (2):

$$4175 = 49 \left(\frac{360 - 49b}{5} \right) + 679b$$

$$4715 = \frac{17640 - 2401b}{5} + 679b$$

$$4715 = 3528 - 480.2b + 679b$$

$$4715 - 3528 = 198.8b$$

$$1187 = 198.8b$$

$$b \approx 5.97$$

Now from (1):

$$360 = 5a + 49(5.94)$$

$$360 = 5a + 292.53$$

$$5a = 67.47 \Rightarrow a \approx 13.49$$

Step 5: Equation of straight line

$$y = a + bX = 13.49 + 5.97X$$

Step 6: Estimate when $X = 15$

$$y = 13.49 + 5.97(15) = 13.49 + 89.55 = 103.44$$

$\therefore$ Estimated Sales = ₹ 103 lakhs when advertising spend is 15 lakhs.

5.3 Review Questions

Short Answer Type Questions

1. Define regression.
2. Define regression analysis.
3. State any two uses of regression analysis.
4. Write the general form of regression equation of Y on X and X on Y
5. State any two differences between correlation and regression.
6. What is a regression line?
7. What is meant by regression coefficients?
8. Define the method of least squares.

Long Answer Type Questions

1. Explain regression and regression analysis with examples.
2. Discuss the uses of regression analysis in business and economics.
3. Explain the types of regression with suitable examples.
4. Explain regression lines and regression equations. Why are there two regression lines?
5. Write the differences between correlation and regression.
6. Explain the method of least squares for fitting a regression line.
7. State the assumptions of Karl Pearson's Co-efficient of Correlation.
8. From the following data fit a straight line trend using least squares method and estimate production for the year 2013.

Production

9. Fit a straight line

Sales ('000)

10. The export

Exports (l)

11. Fit a straight line for advertisement

Advertise

12. A study was made of regression line

Study Hou

Marks

13. The following regression line

Temperat

Sales (l)

14. The experience and estimation

Experienc

Salary (₹)

Years	2006	2007	2008	2009	2010	2011	2012
Production ('000 units)	15	18	17	20	22	24	25

Fit a straight-line trend for the following sales data and estimate the sales for the year 2009.

Years	2000	2001	2002	2003	2004	2005	2006	2007
Sales ('000 units)	12	16	20	25	30	38	40	46

The exports of a company are given below. Fit a straight-line trend and find trend values. Also estimate exports for the year 2015.

Years	2008	2009	2010	2011	2012	2013	2014
Exports (₹ lakhs)	30	28	35	40	45	50	55

Fit a straight line regression by least squares to the following data and estimate sales when advertisement expenditure is ₹12 lakhs.

Advertisement (X in lakhs)	2	4	6	8	10
Sales (Y in lakhs)	20	30	40	50	65

A study was conducted to know whether study hours affect exam performance. Fit a regression line by least squares and predict marks for 7 hours of study.

Study Hours (X)	2	3	4	5	6
Marks (Y)	30	35	45	50	55

The following data shows relationship between temperature and ice-cream sales. Fit a regression line and estimate sales when temperature is 38°C.

Temperature (°C)	20	25	30	35	40
Sales (units)	120	150	180	200	230

The experience of employees and their monthly salary are given below. Fit a regression line and estimate salary for 9 years of experience.

Experience (Years)	1	3	5	7	9	11
Salary (₹ in '000)	12	18	24	28	34	40